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(54) **POWER SUPPLY SYSTEM FOR ARC FURNACE AND ASSOCIATED ARC FURNACE AND METHOD**

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(57) **ABSTRACT**

Provided is a power supply system for arc furnace and a method for controlling the same, and an arc furnace facility having a power supply system, the power supply system includes a power converter, and a polyphase transformer, a first disconnecting means connecting an input of the power converter to a primary circuit of the transformer, and a second disconnecting means connecting an output of the power converter to the primary circuit of the transformer, the power supply system also includes a first isolation switch and a second isolation switch, and a control circuit.

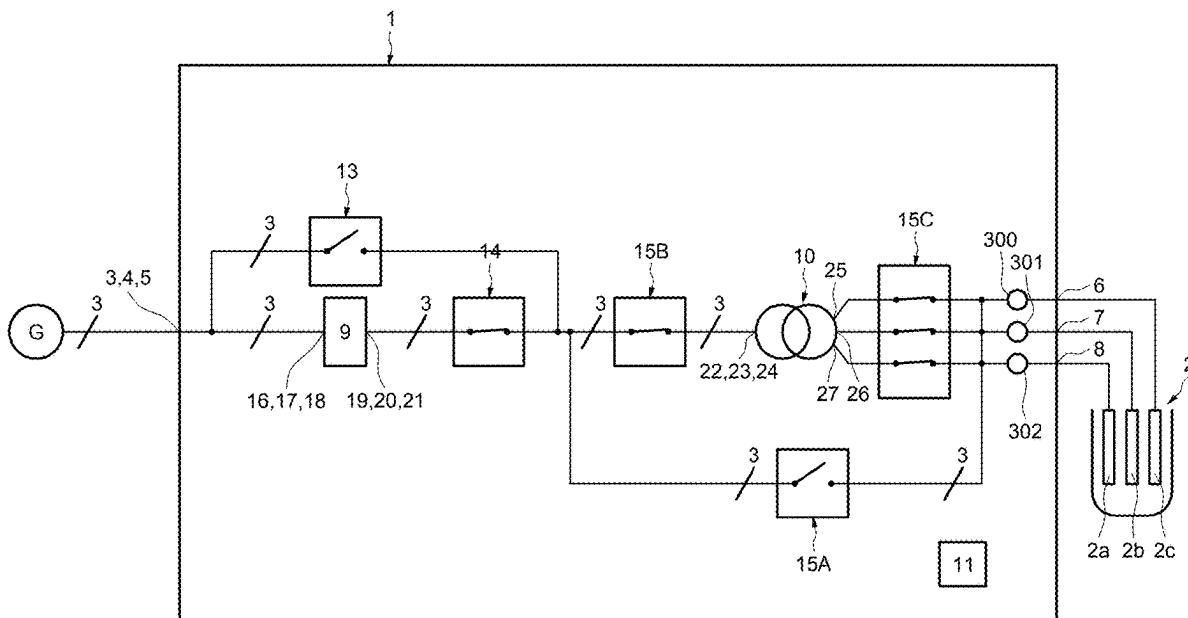


FIG. 1

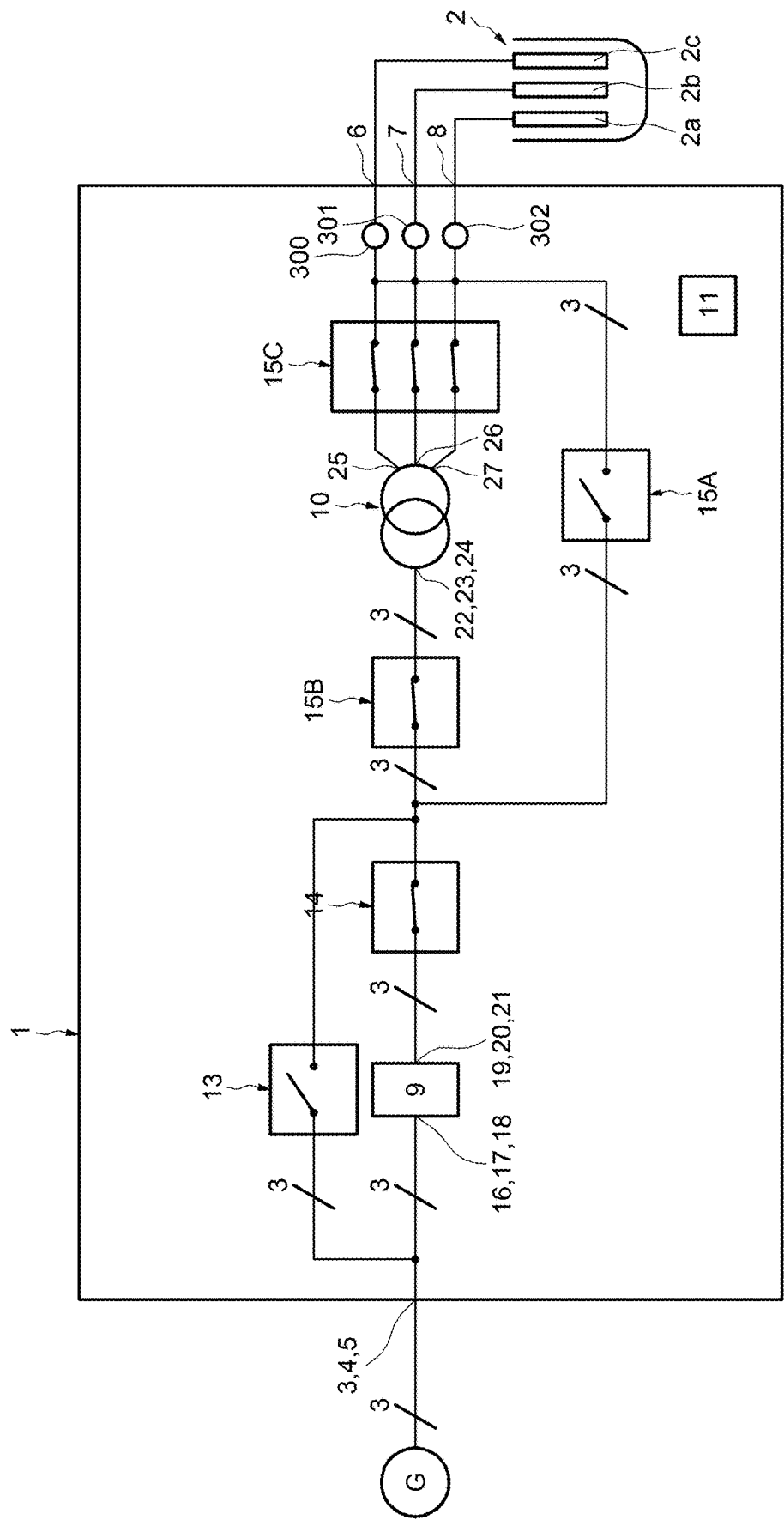


FIG.2

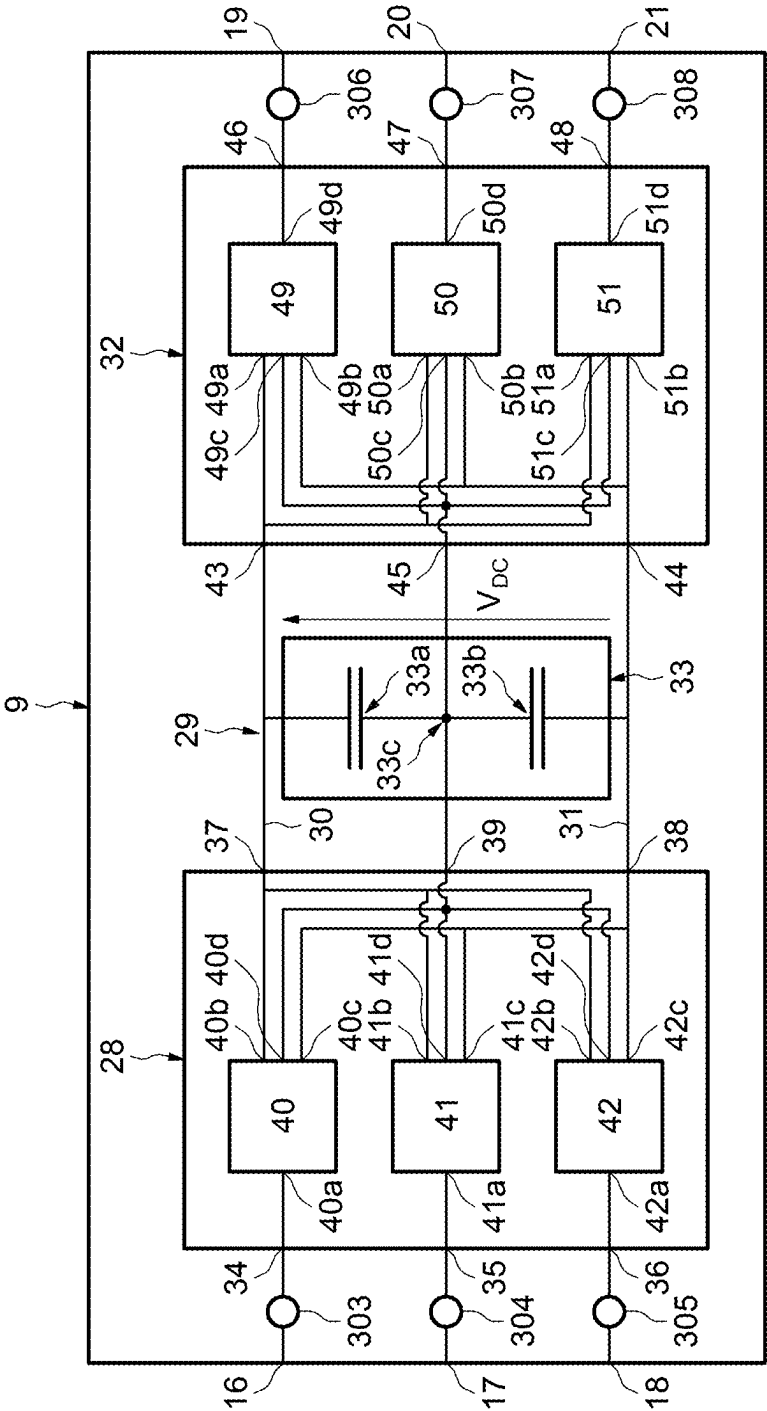


FIG.3

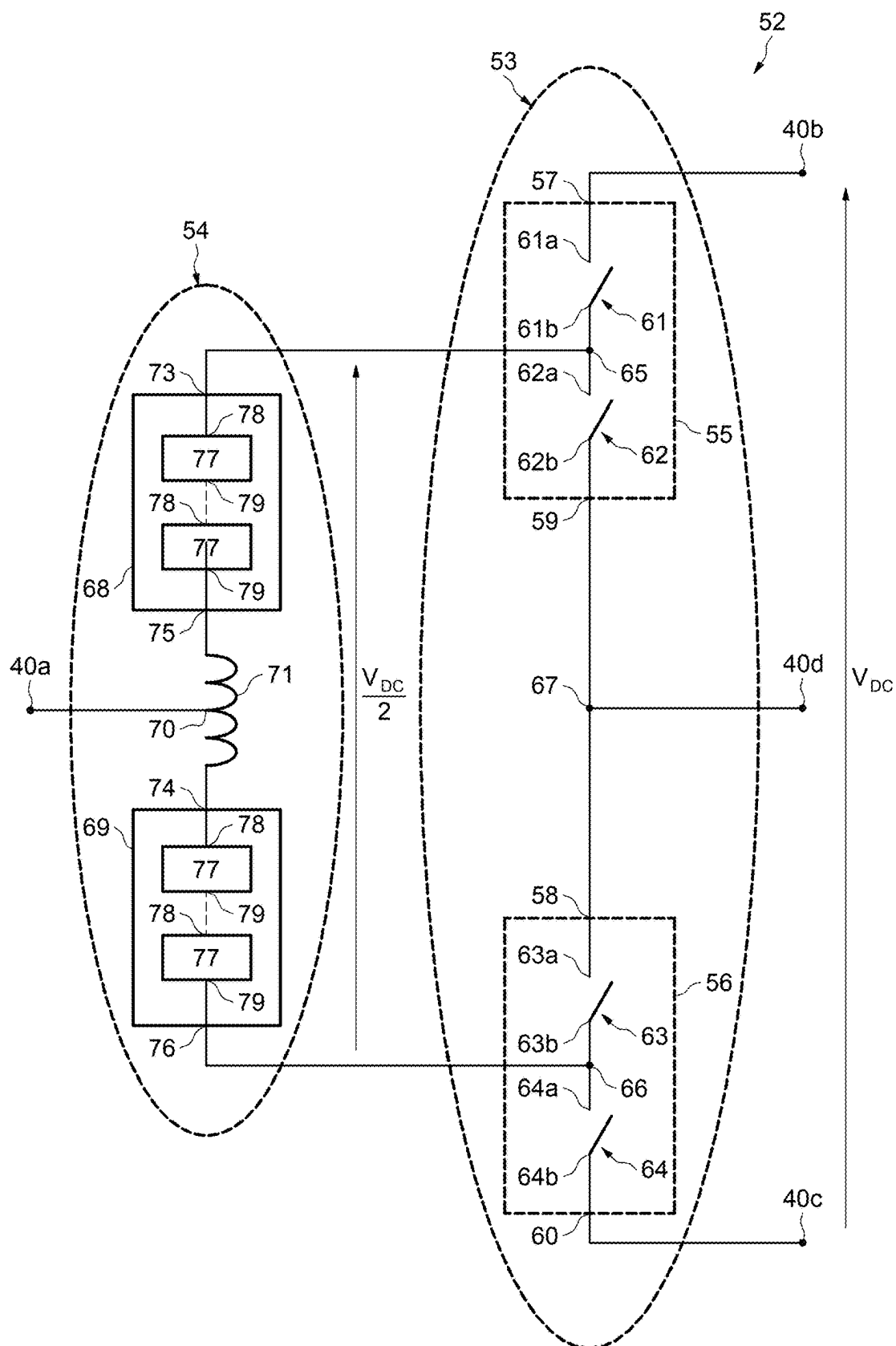


FIG.4

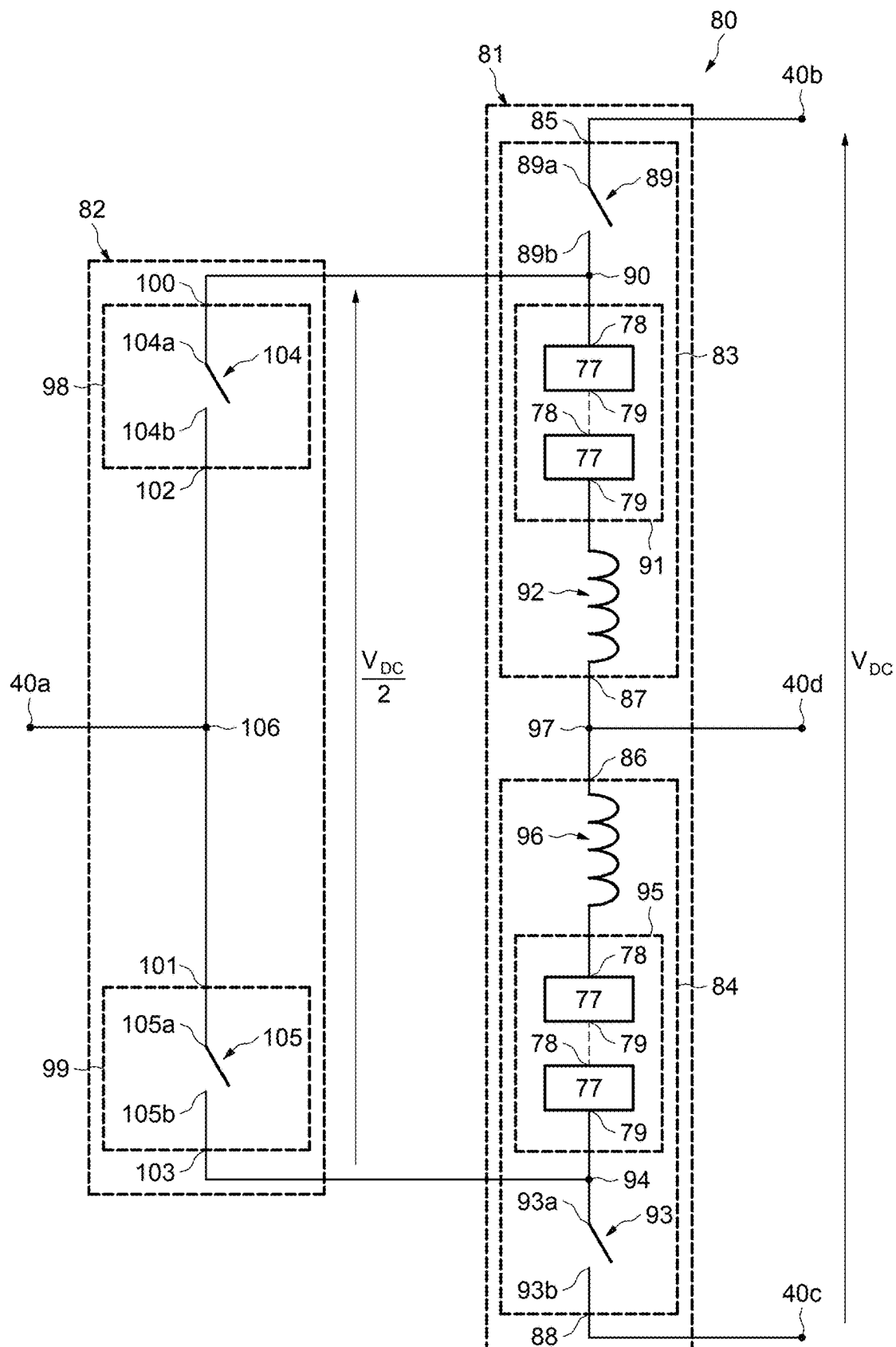


FIG.5

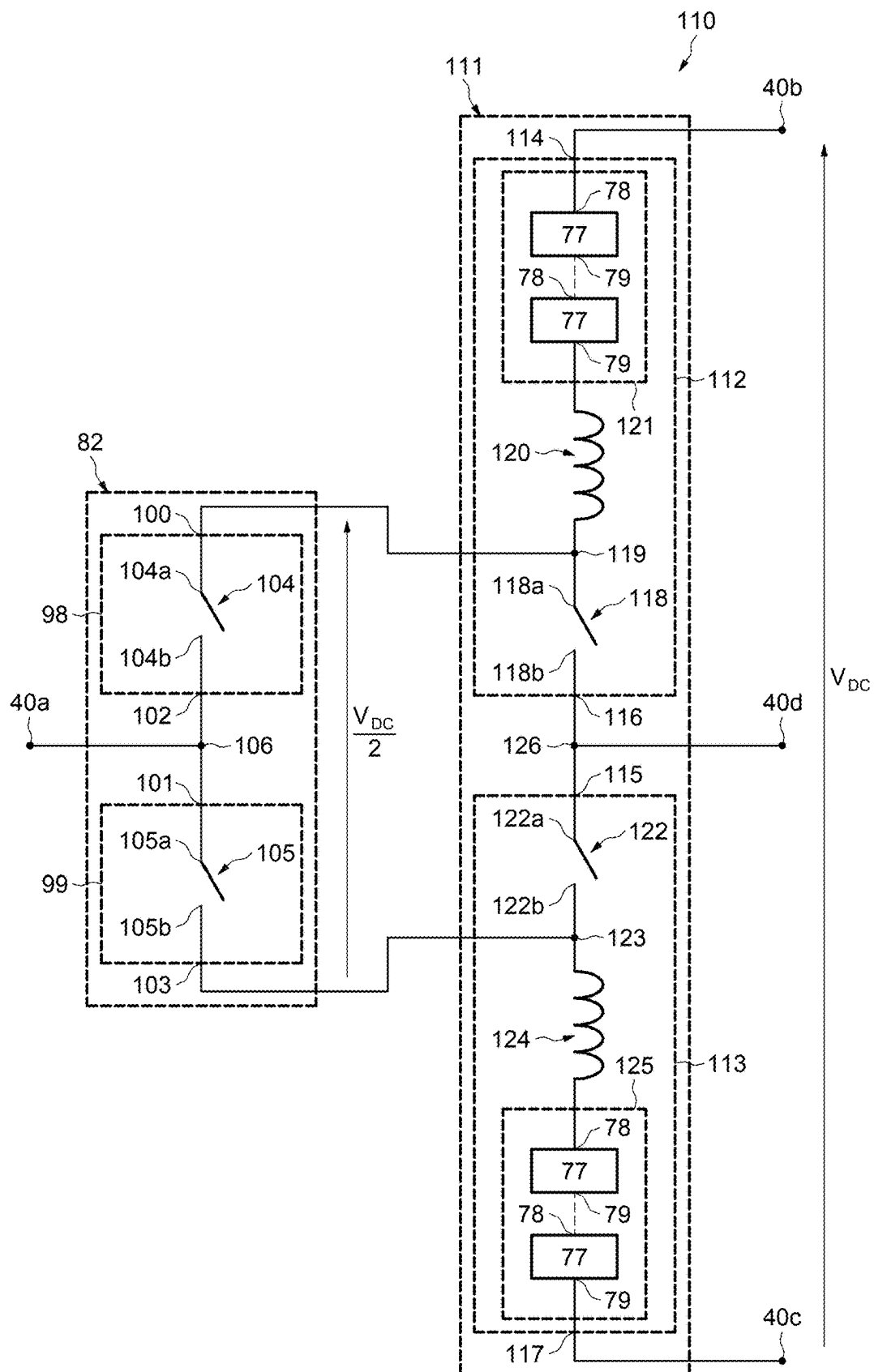


FIG.6

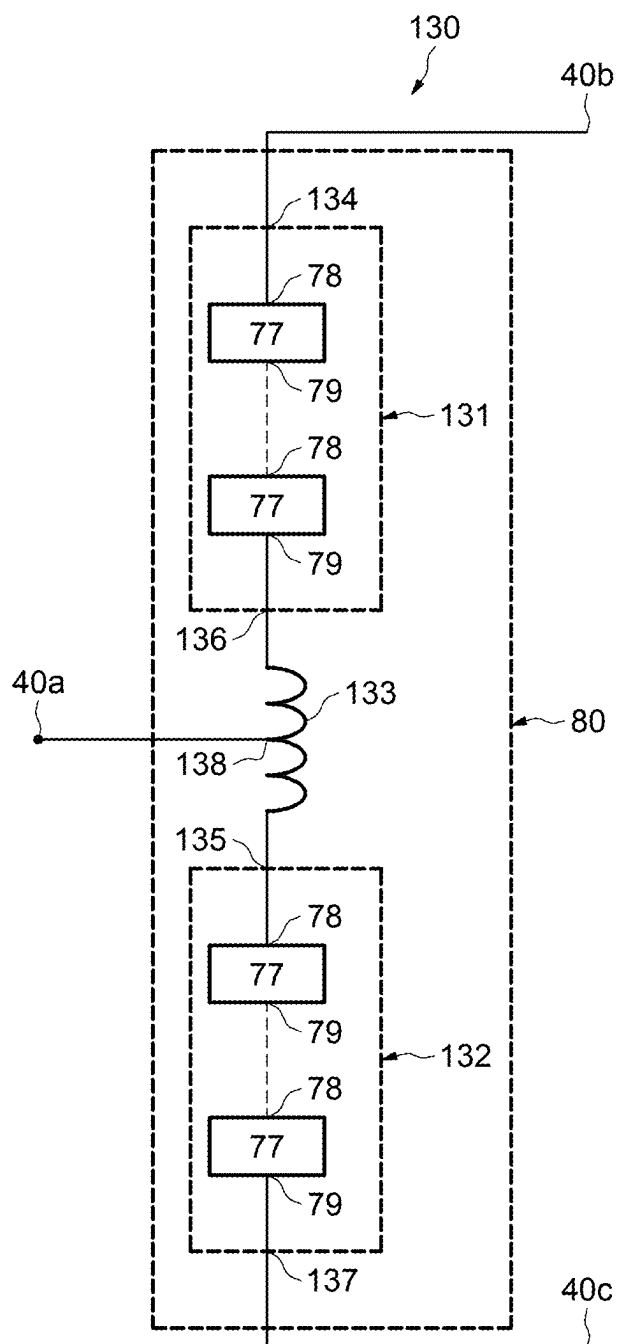


FIG. 7

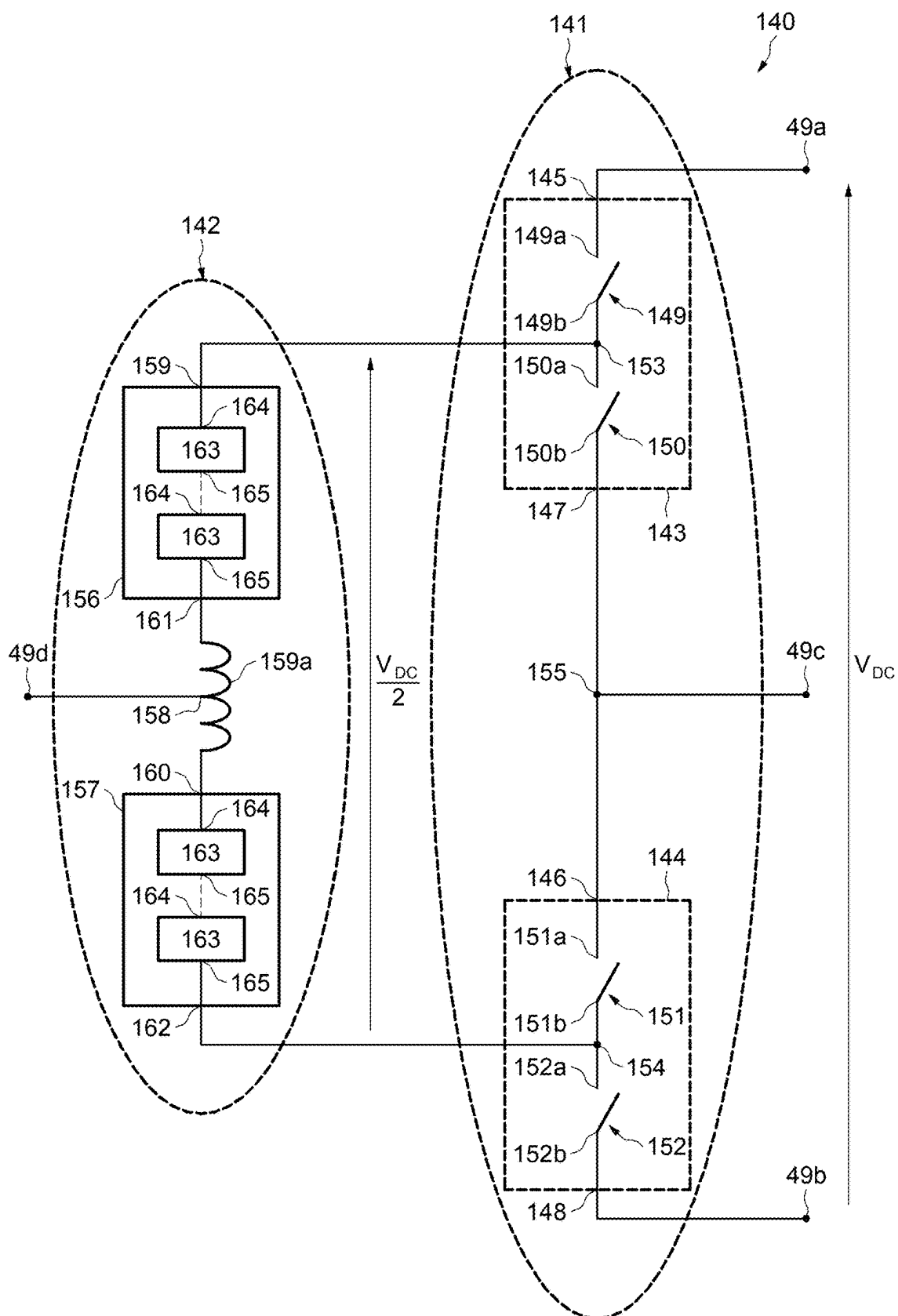


FIG.9

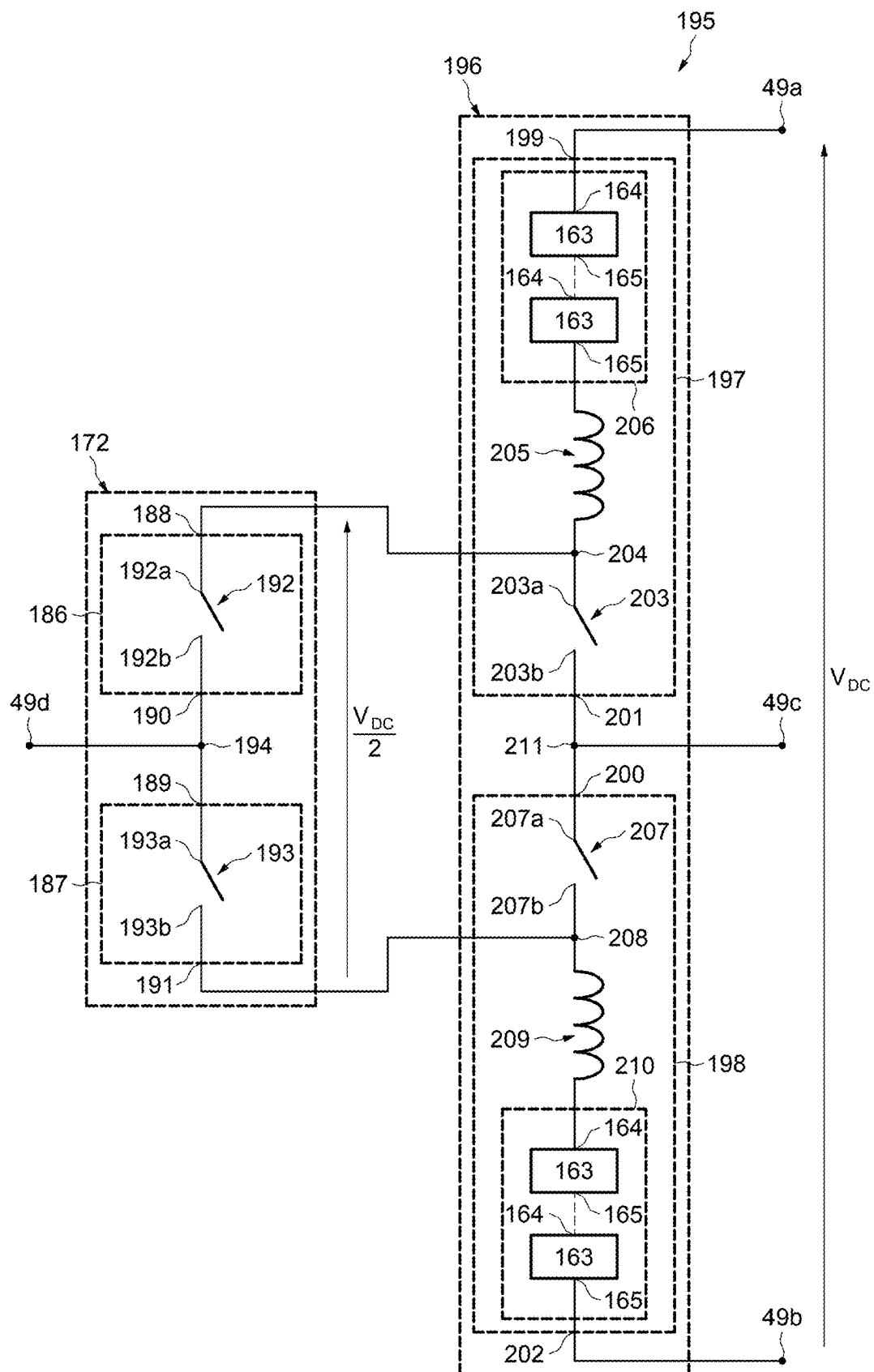


FIG.10

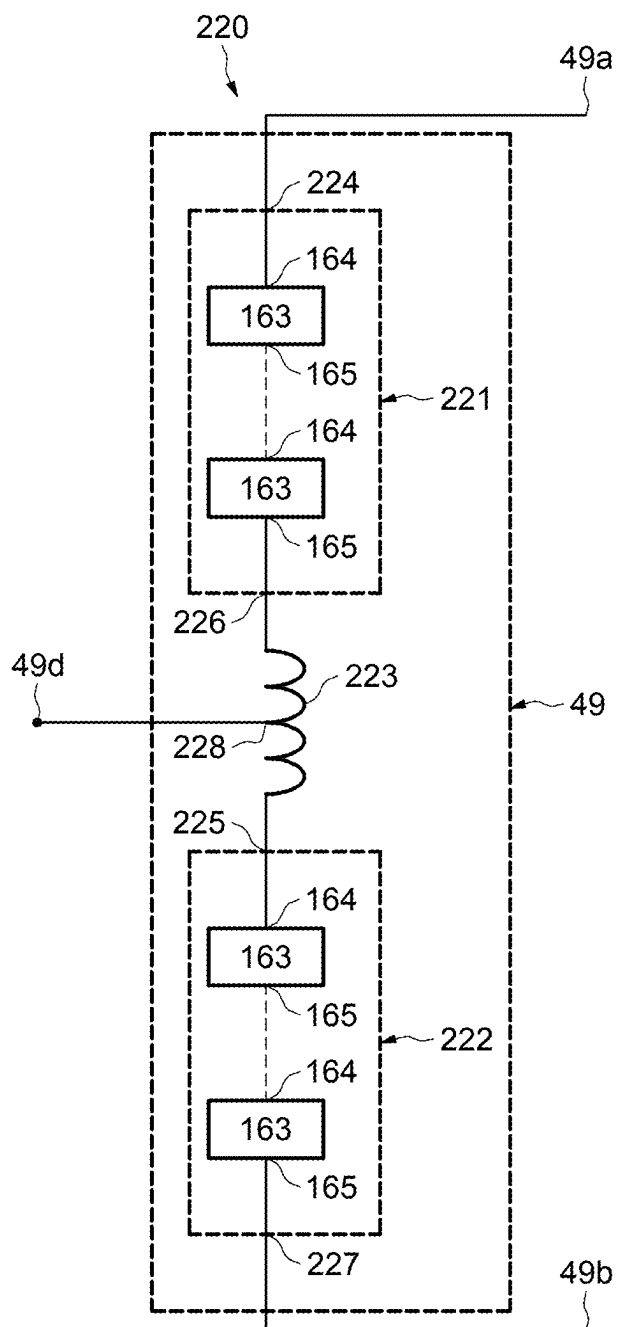


FIG.11

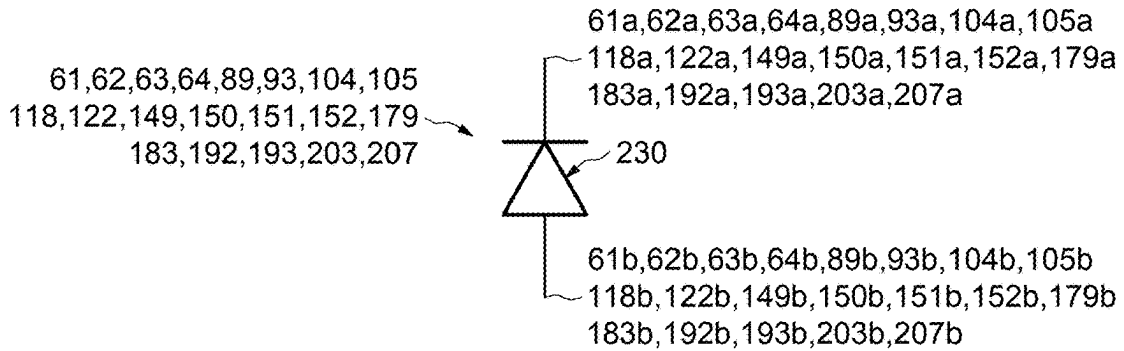


FIG.12

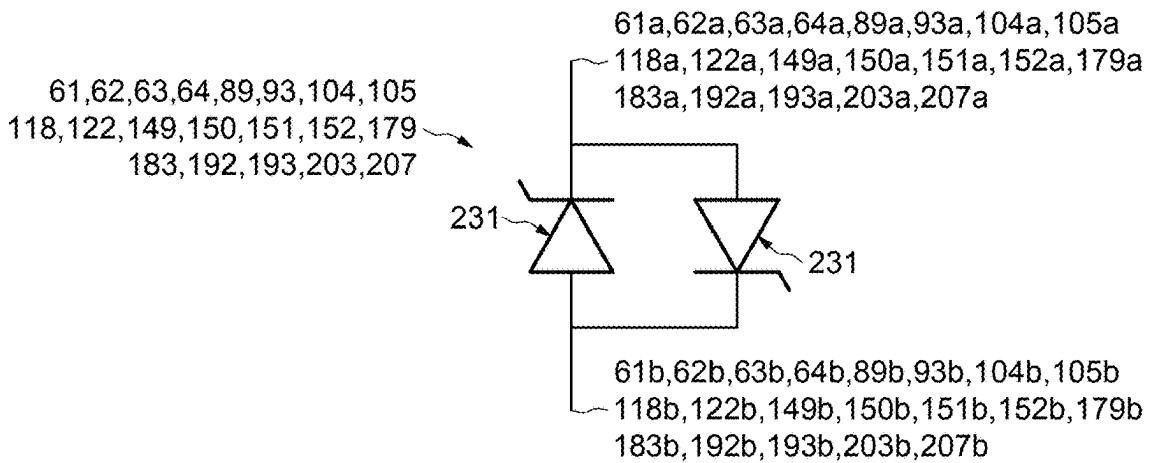


FIG.13

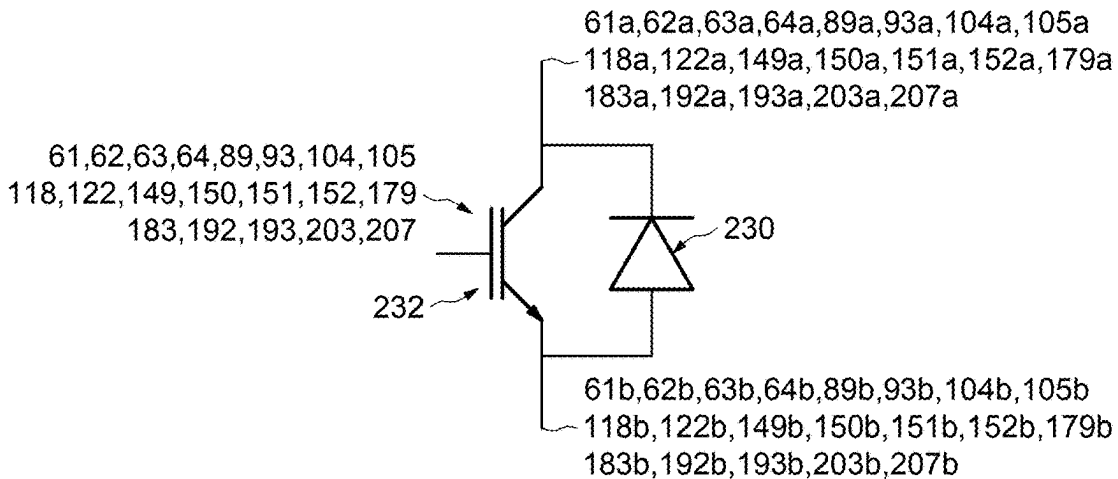


FIG.14

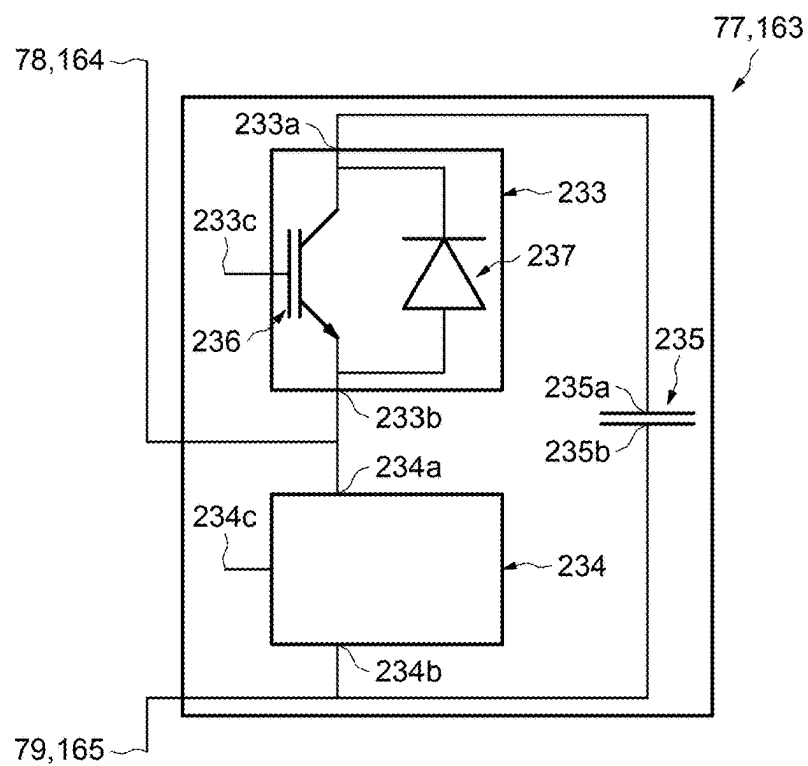


FIG.15

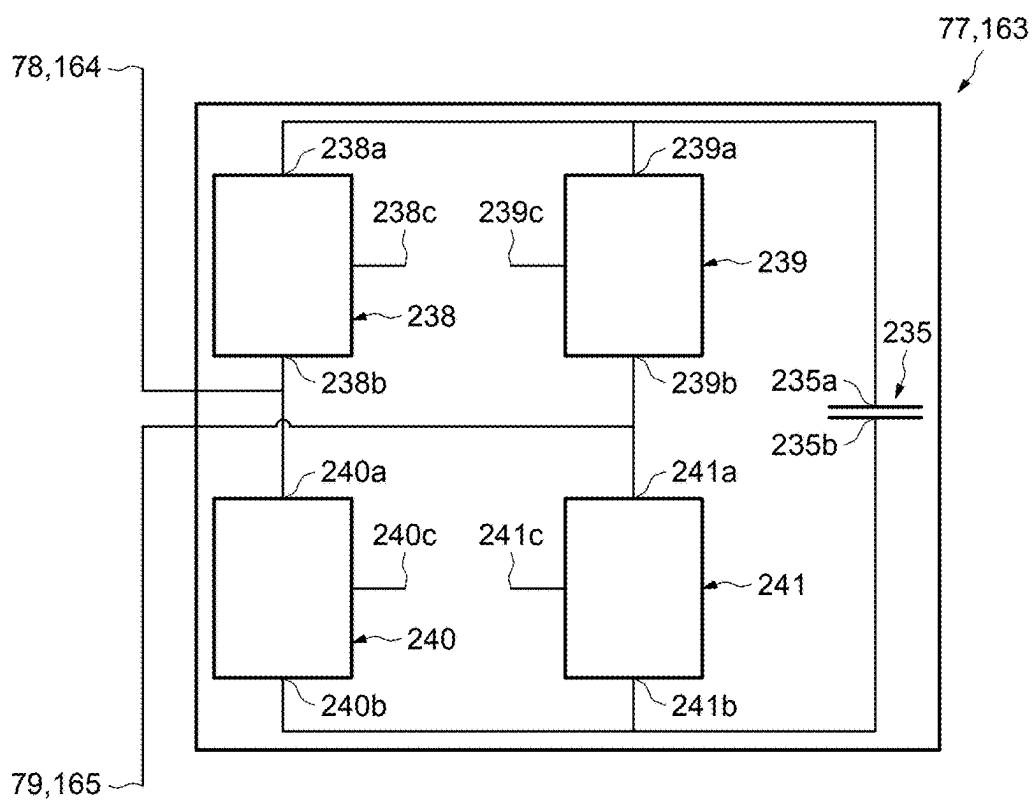
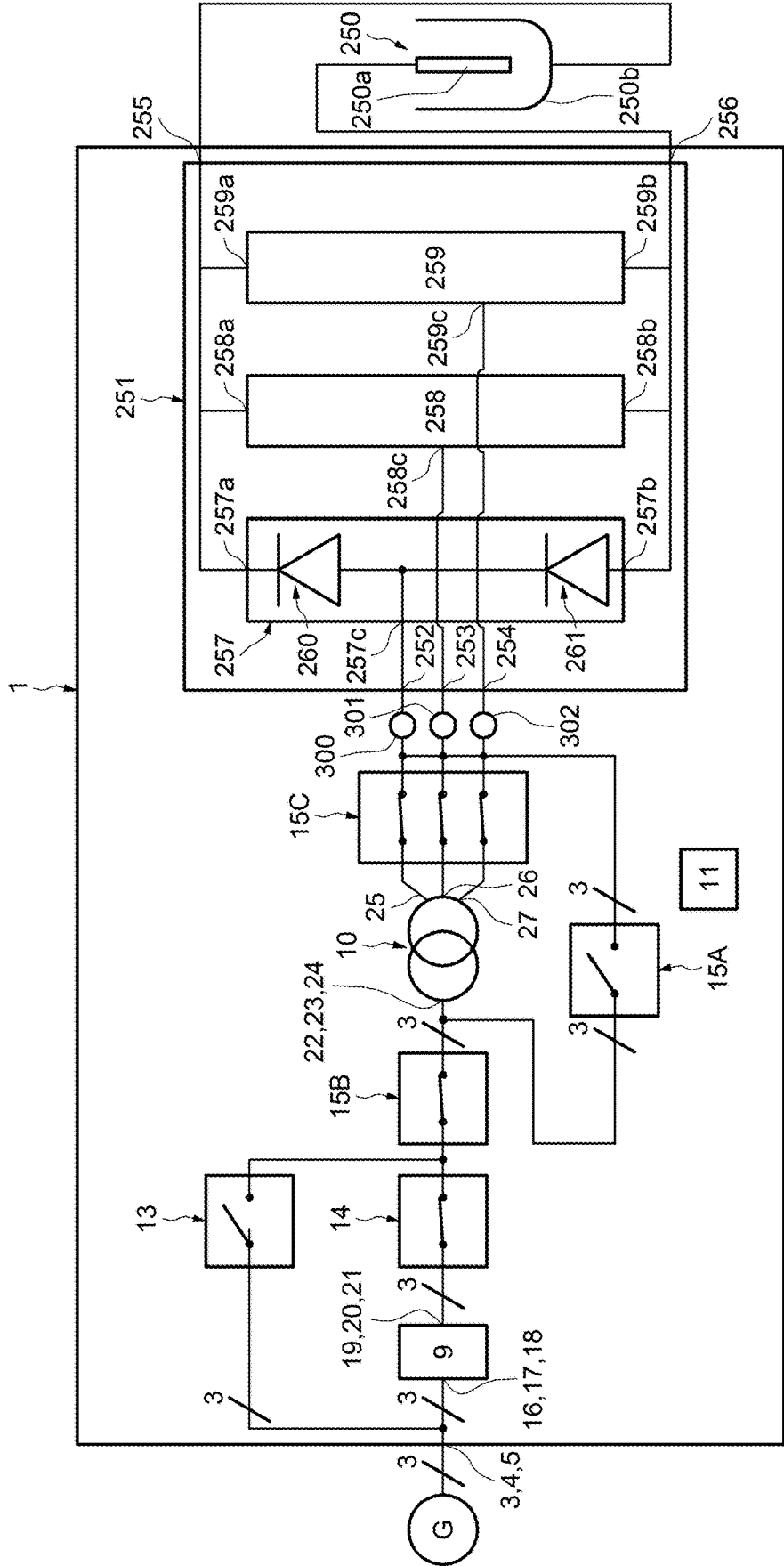


FIG.16



**POWER SUPPLY SYSTEM FOR ARC
FURNACE AND ASSOCIATED ARC
FURNACE AND METHOD**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This application claims priority to European Patent Application Serial Number EP23214191.1, filed Dec. 5, 2023, which is herein incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to arc furnaces, and more particularly a power supply system for arc furnaces, and an arc furnace facility comprising an arc furnace and a power supply system.

[0003] Arc furnaces are highly nonlinear loads resulting in considerable reactions on the power supply system, in particular so-called flicker.

[0004] The document EP 2 329 684 discloses a power supply system to supply a polyphase arc furnace from a grid.

[0005] The power supply system comprises an indirect power converter and a polyphase transformer.

[0006] The indirect power converter supplies a primary circuit of the transformer. A secondary circuit of the transformer supplies electrodes of the arc furnace.

[0007] The indirect power converter is controlled to minimize reactions on the grid in order to suppress the flicker and in order to comply with grid quality standards.

[0008] During the tuning of the power supply system or when the indirect power converter goes wrong, the power supply system may be stop, for example following reactions on the grid generated by the indirect power converter greater than acceptable reactions.

[0009] During these more or less log stops, the arc furnace does not melt scraps to produce molten steel.

[0010] Machines using molten steel, for example rolling mills, are forced to be stopped during stops of the arc furnace generated production losses.

[0011] Further, during normal operation of the arc furnace, when the back-to-back electric arc furnace (EAF) converter is in full operation providing (real) power to the EAF furnace while controlling the voltage and current at the EAF, currents delivered by the indirect power converter going through the transformer generates losses and complicates operation of the arc furnace.

[0012] It is therefore proposed to overcome wholly or partially the disadvantages related to arc furnace power supply systems known from the prior art.

SUMMARY

[0013] In view of the foregoing the invention, provided is a power supply system for arc furnace that includes a power converter comprising an input intended to be connected to a polyphase supply grid and an output, and a polyphase transformer comprising a primary circuit and a secondary circuit, a first disconnecting means, and a second disconnecting means.

[0014] According to an embodiment, the power supply system further comprises a first isolation switch and a second isolation switch, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of

the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch, and a control circuit.

[0015] The control circuit controls the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and to control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed.

[0016] Advantageously, the power converter comprises an input device, a link circuit includes a first bus and a second bus, and an output device, wherein: the input device includes a plurality of input modules each input module being connected to the first bus and the second bus and being intended to be connected to a different phase of the supply grid to supply the link circuit with a continuous voltage from the phase of the supply grid, the output device comprises a plurality of output modules, each output module being connected to the first bus and second bus and to the primary circuit to supply the transformer with an alternative voltage from the circuit link.

[0017] According to an embodiment, each input module comprises a leg that includes a first string having a first set of input switches connected in series, a second set of input switches connected in series, a first connecting node, a second connecting node, and a third connecting node, the first set of input switches operatively coupling the first bus to the third connecting node, and the first connecting node being disposed between two input switches of the first set of input switches, the second set of input switches operatively coupling the second bus to the third connecting node, and the second connecting node being disposed between two input switches of the second set of input switches, a second string having a first set of input switching units in series, a second set of input switching units in series, and a fourth connecting node, the first set of input switching units operatively coupling the first connecting node to the fourth connecting node, the second set of input switching units operatively coupling the second connecting node to the fourth connecting node, and the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit controls the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

[0018] Advantageously, each input module comprises a leg that includes a first string having a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first input switch operatively coupling the first bus to the first connecting node, and the first set of input switching units operatively coupling the first connecting node to the third connecting

node, the second input switch operatively coupling the second bus to the second connecting node, and the second set of input switching units operatively coupling the second connecting node to the third connecting node, a second string comprising: a third input switch, a fourth input switch, and a fourth connecting node, the third input switch operatively coupling the first connecting node to the fourth connecting node, and the fourth input switch operatively coupling the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

[0019] According to an embodiment, each input module comprises a leg that includes a first string having a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first set of input switching units operatively coupling the first bus to the first connecting node, and the first input switch operatively coupling the first connecting node to the third connecting node, the second set of input switching units operatively coupling the second bus to the second connecting node, and the second input switch operatively coupling the second connecting node to the third connecting node, a second string having a third input switch, a fourth input switch, and a fourth connecting node, the third input switch operatively coupling the first connecting node to the fourth connecting node, and the fourth input switch operatively coupling the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit controls the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

[0020] Advantageously, each input module comprises a string that includes a connecting node being intended to be connected to the phase of the supply grid, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first set of input switching units operatively coupling the first bus to the connecting node, and the second set of input switching units operatively coupling the second bus to the connecting node, and the control circuit controls the input switching units, and the output modules.

[0021] According to an embodiment, each output module comprises a leg that includes a first string having a first set of output switches connected in series, a second set of output switches connected in series, a first connecting node, a second connecting node, and a third connecting node, the first set of output switches operatively coupling the first bus to the third connecting node, and the first connecting node being disposed between two output switches of the first set of output switches, the second set of output switches operatively coupling the second bus to the third connecting node, and the second connecting node being disposed between two output switches of the second set of output switches, a second string having a first set of output switching units in series, a second set of output switching units in series, and a fourth connecting node, the first set of output switching units operatively coupling the first connecting node to the third connecting node, the second set of output switching

units operatively coupling the second connecting node to the fourth connecting node, and the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

[0022] Advantageously, each output module comprises a leg that includes a first string having a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first output switch operatively coupling the first bus to the first connecting node, and the first set of output switching units operatively coupling the first connecting node to the third connecting node, the second output switch operatively coupling the second bus to the second connecting node, and the second set of output switching units operatively coupling the second connecting node to the third connecting node, a second string having a third output switch, a fourth output switch, and a fourth connecting node, the third output switch operatively coupling the first connecting node to the fourth connecting node, and the fourth output switch operatively coupling the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

[0023] According to an embodiment, each output module comprises a leg that includes a first string having a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first set of output switching units operatively coupling the first bus to the first connecting node, and the first output switch operatively coupling the first connecting node to the third connecting node, the second set of output switching units operatively coupling the second bus to the second connecting node, and the second output switch operatively coupling the second connecting node to the third connecting node, a second string having a third output switch, a fourth output switch, and a fourth connecting node, the third output switch operatively coupling the first connecting node to the fourth connecting node, and the fourth output switch operatively coupling the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

[0024] Advantageously, each output module comprises a string comprising a connecting node being intended to be connected to a phase of the arc furnace, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first set of output switching units operatively coupling the first bus to the connecting node, and the second set of output switching units operatively coupling the second bus to the connecting node, the control circuit being configured to control the output switching units and the output modules.

[0025] According to an embodiment, the power supply system further comprises a third isolation switch, the third isolation switch connecting the secondary circuit to the second end of the first isolation switch, the control circuit being configured: to control the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second and third isolation switches are closed or the first isolation switch is closed and the second and third isolation switches are open, and to control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second and third isolation switches are closed.

[0026] Another embodiment of the invention relates to an arc furnace facility.

[0027] The arc furnace facility comprises an arc furnace including at least one electrode and a power supply system as defined above, wherein the electrode is connected to at least one phase of the secondary circuit of the polyphase transformer.

[0028] Advantageously, the arc furnace comprises a plurality of electrodes, each electrode being connected to a different phase of the secondary circuit.

[0029] According to an embodiment, the power supply system further comprises a rectifier connected to the electrode and to each phase of the secondary circuit to supply the electrode with a continuous voltage from the secondary circuit.

[0030] Another embodiment of the invention relates to a method for controlling a power supply system of an arc furnace, the power supply system including a power converter comprising an input connected to a polyphase supply grid and an output, and a polyphase transformer comprising a primary circuit and a secondary circuit, first disconnecting means, and second disconnecting means.

[0031] The method comprises controlling the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, a first isolation switch is open and a second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and controlling the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] Other characteristics and advantages of the invention will emerge on reading the following description of embodiments of the invention, provided solely by way of non-limiting examples and with reference to the drawings in which:

[0033] FIG. 1 illustrates schematically a first embodiment of an arc furnace facility according to the invention,

[0034] FIG. 2 illustrates schematically an embodiment of a power converter according to the invention,

[0035] FIG. 3 illustrates schematically a first embodiment of an input module according to the invention,

[0036] FIG. 4 illustrates schematically a second embodiment of the input module according to the invention,

[0037] FIG. 5 illustrates schematically a third embodiment of the input module according to the invention,

[0038] FIG. 6 illustrates schematically a fourth embodiment of the input module known from the prior art,

[0039] FIG. 7 illustrates schematically a first embodiment of an output module according to the invention,

[0040] FIG. 8 illustrates schematically a second embodiment of the output module according to the invention,

[0041] FIG. 9 illustrates schematically a third embodiment of the output module according to the invention,

[0042] FIG. 10 illustrates schematically a fourth embodiment of the output module known from the prior art,

[0043] FIG. 11 illustrates schematically a first embodiment of input switches and output switches according to the invention,

[0044] FIG. 12 illustrates schematically a second embodiment of the input switches and the output switches according to the invention,

[0045] FIG. 13 illustrates schematically a third embodiment of the input switches and the output switches according to the invention,

[0046] FIG. 14 illustrates schematically a first embodiment of input switching units and output switching units according to the invention,

[0047] FIG. 15 illustrates schematically a second embodiment of the input switching units and the output switching units according to the invention, and

[0048] FIG. 16 illustrates a second embodiment of the arc furnace facility according to the invention.

DETAILED DESCRIPTION

[0049] FIG. 1 illustrates a first embodiment of an arc furnace facility.

[0050] The arc furnace facility comprises a polyphase power supply system 1 connected to a grid G and an arc furnace 2 connected to the power supply system 1.

[0051] The power supply system 1 supplies the arc furnace 2 from electrical energy supplied by the grid G.

[0052] It is assumed that the power supply system 1 and the polyphase arc furnace 2 have each three phases, the grid G being a three phases supply grid.

[0053] Each of the power supply system 1 and the polyphase arc furnace 2 may have two phases or more than three phases, the grid G having the same number of phases as the power supply system 1.

[0054] The arc furnace 2 comprises a set of electrodes comprising three electrodes 2a, 2b, 2c.

[0055] The electrodes *2a*, *2b*, *2c* supplied by the power supply system **1** generate electric arcs between the electrodes *2a*, *2b*, *2c* to melt the contents of the arc furnace **2**, for example scrap steel.

[0056] The power supply system **1** comprises three inputs **3**, **4**, **5**, each input being connected to a different phase of the grid **G**, and three outputs **6**, **7**, **8**.

[0057] Each output **6**, **7**, **8** is connected to an electrode *2a*, *2b*, *2c*.

[0058] The power supply system **1** further comprises a power converter **9**, a transformer **10**, a control circuit **11**, current sensors, and voltage sensors.

[0059] The control circuit **11** may comprise a processing unit.

[0060] The power supply system **1** further comprises a first disconnecting means **13** and a second disconnecting means **14**.

[0061] Each disconnecting means of the first and second disconnecting means **13** and **14** comprises for example a disconnecter.

[0062] The power supply system **1** comprises a first isolation switch **15A**, a second isolation switch **15B** and a third isolation switch **15C**.

[0063] The first, second disconnecting means **13**, **14**, and the first, second and third isolation switches **15A**, **15B**, **15C** are controlled by the control circuit **11**.

[0064] The power converter **9** comprises three inputs **16**, **17**, **18** and three outputs **19**, **20**, **21**.

[0065] Each input **16**, **17**, **18** of the power converter **9** is connected to a different input **3**, **4**, **5** of the power supply system **1**.

[0066] The transformer **10** further comprises a primary circuit including three phases **22**, **23**, **24** and a secondary circuit including three phases **25**, **26**, **27**.

[0067] Each phase **25**, **26**, **27** of the secondary circuit is connected to a different output **6**, **7**, **8** of the power supply system **1**.

[0068] Each phase **25**, **26**, **27** of the secondary circuit comprises a sensing module **300**, **301**, **302** comprising a voltage sensor and a current sensor measuring a voltage and a current of the phase.

[0069] The first disconnecting means **13** connects the inputs **16**, **17**, **18** of the power converter **9** to a first end of the first isolation switch **15A**.

[0070] The second disconnecting means **14** connects the outputs **19**, **20**, **21** of the power converter **9** to the first end of the first isolation switch **15A**.

[0071] The second isolation switch **15B** connects the first end of the first isolation switch **15A** to the phases **22**, **23**, **24** of the primary circuit and the third isolation switch **15C** connects the second end of the first isolation switch **15A** to the phases **25**, **26**, **27** of the secondary circuit.

[0072] The first, second and third isolation switch **15A**, **15B**, **15C** are interlocked so that when the first isolation switch **15A** is open, the second and third isolation switches **15B**, **15C** are closed, and when the first isolation switch **15A** is closed, the second and third isolation switches **15B**, **15C** are open. The first isolation switch **15A** is normally open whereas the second and third isolation switches are normally closed.

[0073] In another embodiment, the power supply system **1** does not comprise the third isolation switch **15C**, the second end of the first isolation switch **15A** being connected to the phases **25**, **26**, **27** of the secondary circuit.

[0074] FIG. **2** illustrates an embodiment of the power converter **9**.

[0075] The power converter **9** further comprises an input device **28**, a link circuit **29** comprising a first bus **30** and a second bus **31**, and an output device **32**.

[0076] Each input **16**, **17**, **18** of the power converter **9** comprises a sensing module **303**, **304**, **305** comprising a voltage sensor and a current sensor measuring a voltage and a current of the input.

[0077] Each output **19**, **20**, **21** of the power converter **9** comprises a sensing module **306**, **307**, **308** comprising a voltage sensor and a current sensor measuring a voltage and a current of the output.

[0078] According to one embodiment, the first bus **30** may have a potential higher than the second bus **31**.

[0079] The voltage between the first and second buses **30**, **31** is labeled VDC in the figures.

[0080] The link circuit **29** may further comprise a direct current DC link module **33** including two energy storage devices **33a**, **33b** connected in series.

[0081] A first end of a first energy storage device **33a** is connected to the first bus **30**, a first end of the second energy storage **33b** is connected to the second bus **31**, and the second end of the second energy storage **33b** and the second end of the first energy storage **33a** are connected to a connecting node **33c**.

[0082] Each energy storage device **33a**, **33b** comprises for example a capacitor.

[0083] The input device **28** comprises three inputs **34**, **35**, **36** and three outputs **37**, **38**, **39**.

[0084] Each input **34**, **35**, **36** of the input device **28** is connected to a different input **16**, **17**, **18** of the power converter **9**.

[0085] A first output **37** of the input device **28** is connected to the first bus **30**, a second output **38** of the input device **28** is connected to the second bus **31** and the third output **39** is connected to the connecting node **33c** of the DC link module **33**.

[0086] The input device **28** further comprises identical input modules **40**, **41**, **42** intended to supply the link circuit **29** with a continuous voltage from the phases of the supply grid.

[0087] Each input module **40**, **41**, **42** is connected to a different input **34**, **35**, **36** of the input device **28**.

[0088] Each input module **40**, **41**, **42** comprises an input **40a**, **41a**, **42a** connected to the input **34**, **35**, **36** of the input device **28**, a first output **40b**, **41b**, **42b** connected to the first output **37** of the input device **28**, a second output **40c**, **41c**, **42c** connected to the second output **38** of the input device **28**, and a third output **40d**, **41d**, **42d** connected to the third output **39** of the input device **28**.

[0089] The output device **32** comprises three inputs **43**, **44**, **45** and three outputs **46**, **47**, **48**.

[0090] A first input **43** of the output device **32** is connected to the first bus **30**, a second input **44** of the output device **32** is connected to the second bus **31**, and the third input **45** of the output device **32** is connected to the connecting node **33c** of the DC link module **33**.

[0091] A first output **46** of the output device **32** is connected to a first output **19** of the power converter **9**, a second output **47** is connected to a second output **20** of the power converter **9**, and a third output **48** is connected to the third output **21** of the power converter **9**.

[0092] The output device 32 further comprises identical output modules 49, 50, 51 intended to supply the transformer 10 with alternative voltages from the link circuit 29.

[0093] Each output module 49, 50, 51 comprises a first input 49a, 50a, 51a connected to the first input 43 of the output device 32, a second input 49b, 50b, 51b connected to the second input 44 of the output device 32 and a third input 49c, 50c, 51c connected to the third input 45 of the output device 32.

[0094] Each output module 49, 50, 51 further comprises an output 49d, 50d, 51d connected to a different output 46, 47, 48 of the output device 32.

[0095] The input device 28 is intended to supply the link circuit 29 with a continuous voltage from the grid G.

[0096] The output device 32 is intended to supply with electrical energy the electrodes 2a, 2b, 2c through the transformer 10.

[0097] The energy storage devices 33a, 33b are designed to stabilize the continuous voltage and to smooth the current variation on the buses 30, 31.

[0098] The DC link module 33 may decouple the input device 28 and the output device 32 to avoid that electrical perturbations generated by the arc furnace 2 are rejected on the grid G.

[0099] In another embodiment, the link circuit 29 may not comprise the DC link module 33, the third output 39 of the input device 28 and the third input 45 of the output device 32 being not connected.

[0100] As the input modules 40, 41, 42 are identical, FIG. 3 illustrates schematically a first embodiment of the input module 40 comprising a first embodiment of a hybrid modular multilevel converter (HMMC).

[0101] The first embodiment of HMMC comprises a leg 52 comprising a first string 53 and a second string 54.

[0102] The first string 53 comprises a first branch 55 and a second branch 56.

[0103] Each branch 55, 56 comprises a first end 57, 58 and a second end 59, 60.

[0104] The first end 57 of the first branch 55 is connected to the first output 40b of the input module 40 and the second end 60 of the second branch 56 is connected to the second output 40c of the input module 40.

[0105] Each branch 55, 56 comprises a first set of input switches 61, 62, 63, 64 connected in series between the first and the second ends 57, 58, 59, 60 of the branch 55, 56.

[0106] The first branch 55 further comprises a first connecting node 65 situated between two input switches 61, 62 of the first set of input switches.

[0107] The second branch 56 further comprises a second connecting node 66 situated between two input switches 63, 64 of the second set of input switches.

[0108] As represented, the number of input switches between the first connecting node 65 and the first end 57 of the branch 55, 56 may equal to the number of input switches between the first connecting node 65 and the second end 59, and the number of input switches between the second connecting node 66 and the first end 58 of the branch 55, 56 may equal to the number of input switches between the second connecting node 66 and the second end 60.

[0109] Each branch 55, 56 may comprise more than two input switches, the two branches 55, 56 comprising a same number of input switches.

[0110] Each input switch 61, 62, 63, 64 comprises a first end 61a, 62a, 63a, 64a and a second end 61b, 62b, 63b, 64b.

[0111] The input switches 61, 62 of the first branch 55 are connected in series so that the first end 61a of the input switch 61 is connected to the first end 57 of the first branch 55, the second end 61b and the first end 62a of the input switches 61, 62 are connected to the first connecting node 65, and the second end 62b of the input switch 62 is connected to the second end 59 of the first branch 55.

[0112] The input switches 63, 64 of the second branch 56 are connected in series so that the first end 63a of the input switch 63 is connected to the first end 58 of the second branch 56, the second end 63b and the first end 64a of the input switches 63, 64 are connected to the second connecting node 66, and the second end 64b of the input switch 64 is connected to the second end 60 of the second branch 56.

[0113] The first string 53 further comprises a third connecting node 67 connected to the second end 59 of the branch 55, the first end 58 of the second branch 56, and the third output 40d of the input module 40.

[0114] The first set of input switches 61, 62 operatively couple the first bus 30 to the third connecting node 67, and the second set of input switches 63, 64 operatively couple the second bus 31 to the third connecting node 67.

[0115] The second string 54 comprises a first portion 68, a second portion 69, and a fourth connecting node 70.

[0116] The second string 54 may further comprise an inductor 71.

[0117] Each portion 68, 69 comprises a first end 73, 74, a second end 75, 76.

[0118] The first portion 68 comprises a first set of input switching units 77 in series between the first and second ends 73, 75 of the portion 68, 69.

[0119] The second portion 69 comprises a second set of input switching units 77 in series between the first and second ends 74, 76 of the portion 68, 69.

[0120] Each set of input switching units 77 comprises a same number of input switching units 77.

[0121] Each input switching unit 77 includes a plurality of controllable switches and energy storage means, the controllable switches being intended to bridge the energy storage means or to connect the energy storage means in each input switching unit 77.

[0122] The input switching units 77 of each set are connected in series so that the first connection 78 of an input switching unit 77 is connected to a second connection 79 of an adjacent input switching unit 77.

[0123] The first remaining connection 78 of the switching unit 77 of the first set is connected to the first end 73 and the second remaining connection 79 of the input switching unit 77 of the first set is connected to the second end 75.

[0124] The first remaining connection of an input switching unit 77 is the first connection 78 of the input switching unit which is not connected to the first connection 78 or to the second connection 79 of another input switching unit 77.

[0125] The second remaining connection 79 of an input switching unit 77 is the second connection 79 of the input switching unit which is not connected to the first connection 78 or to the second connection 79 of another input switching unit 77.

[0126] The first end 73 of the first portion 68 is connected to the first connecting node 65 and the second end 75 is connected to a first end of the inductor 71.

[0127] The first remaining connection 78 of the input switching unit 77 of the second set is connected to the first

end 74 and the second remaining connection 79 of the input switching unit 77 of the second set is connected to the second end 76.

[0128] The first set of input switching units 77 operatively couples the first connecting node 66 to the fourth connecting node 70 and the second set of input switching units 77 operatively couples the second connecting node 66 to the fourth connecting node 70.

[0129] The first end 74 of the second portion 69 is connected to a second end of the inductor 71 and the second end 76 is connected to the second connecting node 66.

[0130] The fourth connecting node 70 connects the input 40a of the input module 40 to a mid-connecting node of the inductor 71.

[0131] The inductor 71 permits to limit current peaks in the second string 54 and the phases of the grid G during commutations of the input switching units 77.

[0132] In another embodiment, the second string 54 does not comprise the inductor 71, the ends 74, 75 of the first and second portions 68, 69 and the input 40a of the input module 40 being connected to the fourth connecting node 70, the first portion 68 or the second portion 69 comprising an inductor to limit the current peaks in the second string 54 and the second string 54 comprises another inductor connecting the input 40a of the input module 40 to the fourth connecting node 70 to limit the current peaks in the phase of the grid G.

[0133] In another embodiment, the second string 54 does not comprise the inductor 71, the ends 74, 75 of the first and second portions 68, 69 and the input 40a of the input module 40 are connected to the fourth connecting node 70, each of the first portion 68 and the second portion 69 comprising an inductor to limit the current peaks in the second string 54 and the second string 54 may comprise another inductor connecting the input 40a of the input module 40 to the fourth connecting node 70 to limit the current peaks in the phase of the grid G.

[0134] FIG. 4 illustrates schematically a second embodiment of the input module 40 comprising a second embodiment of the HMMC.

[0135] The second embodiment of HMMC comprises a leg 80 comprising a first string 81 and a second string 82.

[0136] The first string 81 comprises a first branch 83 and a second branch 84.

[0137] Each branch 83, 84 comprises a first end 85, 86 and a second end 87, 88.

[0138] The first end 85 of the first branch 83 is connected to the first output 40b of the input module 40 and the second end 88 of the second branch 84 is connected to the second output 40c of the input module 40.

[0139] The first branch 83 comprises a first input switch 89, a first connecting node 90, a first set 91 of input switching units 77 connected in series, and a first inductor 92.

[0140] A first end 89a of the first input switch 89 is connected to the first end 85 of the first branch 83 and a second end 89b of the first input switch 89 is connected to the first connecting node 90.

[0141] The input switching units 77 of the first set 91 are connected in series so that the first connection 78 of an input switching unit 77 is connected to a second connection 79 of an adjacent input switching unit 77.

[0142] The first remaining connection 78 of the input switching unit 77 of the first set 91 is connected to the first

connecting node 90 and the second remaining connection 79 of the input switching unit 77 of the first set 91 is connected to a first end of the first inductor 92.

[0143] The second end of the first inductor 92 is connected to the second end 87 of the first branch 83.

[0144] The second branch 84 comprises a second input switch 93, a second connecting node 94, a second set 95 of input switching units 77 connected in series, and a second inductor 96.

[0145] A second end 93b of the second input switch 93 is connected to the second end 88 of the second branch 84 and a first end 93a of the second input switch 93 is connected to the second connecting node 94.

[0146] The input switching units 77 of the second set 95 are connected in series so that the first connection 78 of an input switching unit 77 is connected to a second connection 79 of an adjacent input switching unit 77.

[0147] The second remaining connection 79 of the input switching unit 77 of the second set 95 is connected to the second connecting node 94 and the first remaining connection 78 of the input switching unit 77 of the second set 95 is connected to a first end of the second inductor 96.

[0148] The second end of the second inductor 96 is connected to the first end 86 of the second branch 84.

[0149] The first and second sets 91, 95 of input switching units 77 have a same number of input switching units 77.

[0150] The first string 81 further comprises third connecting node 97 connected to the first end 86 of the second branch 84, the second end of the first branch 83, and the third input 40d of the input module 40.

[0151] The first input switch 89 connects the first bus 30 to the first connecting node 90 and the input switching units 77 connects in series of the first set 91 of input switching units 77.

[0152] The input switching units 77 connected in series of the first set 91 of input switching units 77 connected in series with the first inductor 92 connect the first connecting node 90 to the third connecting node 97.

[0153] The second input switch 93 connects the second bus 31 to the second connecting node 94 and the second set 95 of input switching units 77.

[0154] The input switching units 77 connected in series of the second set 95 of input switching units 77 connected in series with the second inductor 96 connect the second connecting node 94 to the third connecting node 97.

[0155] Each branch 83, 84 may comprise more than one input switch, the input switches being connected in series between a connecting node and one end of the branch.

[0156] The first and second inductors 92, 96 limit current peaks in the first string 81 and the phases of the grid G during commutations of the input switching units 77.

[0157] The second string 82 comprises a first portion 98 and a second portion 99.

[0158] Each portion 98, 99 comprises a first end 100, 101 and a second end 102, 103.

[0159] The first portion 98 comprises a third input switch 104 comprising a first end 104a connected to the first end 100 of the portion 98, 99 and a second end 104b connected to the second end 102 of the portion 98, 99.

[0160] The second portion 99 comprises a fourth input switch 105 comprising a first end 105a connected to the first end 101 of the portion 98, 99 and a second end 105b connected to the second end 103 of the portion 98, 99.

[0161] In a variant, each portion 98, 99 may comprise more than one input switch connected in series between both ends of the portion 98, 99, the number of input switches of the first and second portions 98, 99 being identical.

[0162] The second string 82 further comprises a fourth connecting node 106 connected to the second end 102 of the first portion 98, the first end 101 of the second portion 99, and the input 40a of the input module 40.

[0163] The first end 100 of the first portion 98 is connected to the first connecting node 90 and the second end 103 of the second portion 99 is connected to the second connecting node 94.

[0164] The third input switch 104 operatively couples the first connecting node 90 to the fourth connecting node 106 and the fourth input switch 105 operatively couples the second connecting node 94 to the fourth connecting node 106.

[0165] FIG. 5 illustrates schematically a third example of the input module 40 comprising a third example of the HMMC.

[0166] The third example of HMMC comprises a leg 110 comprising a first string 111 and the second string 82 according to the second example of HMMC illustrated in FIG. 4.

[0167] The first string 111 comprises a first branch 112 and a second branch 113.

[0168] Each branch 112, 113 comprises a first end 114, 115 and a second end 116, 117.

[0169] The first end 114 of the first branch 112 is connected to the first output 40b of the input module 40 and the second end 117 of the second branch 113 is connected to the second output 40c of the input module 40.

[0170] The first branch 112 comprises a first input switch 118, a first connecting node 119, a first inductor 120, and a first set 121 of input switching units 77 connected in series.

[0171] A second end 118b of the first input switch 118 is connected to the second end 116 of the first branch 112 and a first end 118a of the first input switch 118 is connected to the first connecting node 119.

[0172] The input switching units 77 of the first set 121 are connected in series so that the first connection 78 of an input switching unit 77 is connected to a second connection 79 of an adjacent input switching unit 77.

[0173] The first remaining connection 78 of the input switching unit 77 of the first set 121 is connected to the first end 114 of the first branch 112 and the second remaining connection 79 of the input switching unit 77 of the first set 121 is connected to a first end of the first inductor 120.

[0174] The second end of the first inductor 120 is connected to the first connecting node 119.

[0175] The second branch 113 comprises a second input switch 122, a second connecting node 123, a second inductor 124, and a second set 125 of input switching units 77 connected in series.

[0176] A first end 122a of the second input switch 122 is connected to the first end 115 of the second branch 113 and a second end 122b of the second input switch 122 is connected to the second connecting node 123.

[0177] The input switching units 77 of the second set 125 are connected in series so that the first connection 78 of an input switching unit 77 is connected to a second connection 79 of an adjacent input switching unit 77.

[0178] The first remaining connection 78 of the input switching unit 77 of the second set 125 is connected to a first end of the second inductor 124.

[0179] A second end of the second inductor 124 is connected to the second connecting node 123.

[0180] The second remaining connection 79 of the input switching unit 77 of the second set 125 is connected to the second end 117 of the second branch 113.

[0181] The first and second sets 121, 125 of input switching units 77 have a same number of input switching units 77.

[0182] The first string 111 further comprises third connecting node 126 connected to the first end 115 of the second branch 113, the second end 116 of the first branch 112, and the third input 40d of the input module 40.

[0183] The first set 121 of input switching units 77 operatively couples the first bus 30 to the first connecting node 119 and the first input switch 118 operatively coupling the first connecting node 119 to the third connecting node 126.

[0184] The second set 125 of input switching units 77 operatively couples the second bus 31 to the second connecting node 123 and the second input switch 122 operatively couples the second connecting node 123 to the third connecting node 126.

[0185] Each branch 112, 113 may comprise more than one input switch, the input switches being connected in series between a connecting node and one end of the branch.

[0186] The first and second inductors 120, 124 limit current peaks in the first string 111 and the phases of the grid G during commutations of the input switching units 77.

[0187] The first end 100 of the first portion 98 of the second string 82 is connected to the first connecting node 119 and the second end 103 of the second portion 99 of the second string 82 is connected to the second connecting node 123.

[0188] FIG. 6 illustrates schematically a fourth embodiment of the input module 40 comprising an embodiment of a modular multi-level converter MMC.

[0189] The output module 40 comprises a string 130 including a first portion 131 and a second portion 132.

[0190] The input module 40 comprises an inductor 133.

[0191] Each portion 131, 132 comprises a first end 134, 135 and a second end 136, 137.

[0192] The first end 134 of the first portion 131 is connected to the first output 40b of the input module 40 and the second end 137 of the second portion 132 is connected to the second output 40c of the input module 40.

[0193] The third output 40d of the input module 40 is not connected.

[0194] The first portion 131 comprises a first set of input switching units 77 in series between the first and second ends 134, 136 of the portion 131, 132.

[0195] The second portion 132 comprises a second set of input switching units 77 in series between the first and second ends 135, 137 of the portion 131, 132.

[0196] Each set of input switching units 77 comprises a same number of switching units 77.

[0197] The input switching units 77 of each set are connected in series so that the first connection 78 of an input switching unit 77 is connected to a second connection 79 of an adjacent input switching unit 77.

[0198] The first remaining connection 78 of the input switching unit 77 of the first set is connected to the first end

134 and the second remaining connection **79** of the input switching unit **77** of the first set is connected to the second end **136**.

[0199] The first remaining connection **78** of the input switching unit **77** of the second set is connected to the first end **135** and the second remaining connection **79** of the input switching unit **77** of the second set is connected to the second end **137**.

[0200] The string **130** further comprises a connecting node **138**.

[0201] The second end **136** of the first portion **131** is connected to a first end of the inductor **133** and the first end **135** of the second portion **132** is connected to the second end of the inductor **133**.

[0202] The connecting node **138** connects the input **40a** of the input module **40** to a mid-connecting node of the inductor **133**.

[0203] The first set of input switching units **77** operatively couples the first bus **30** to the connecting node **138** and the second set of input switching units **77** operatively couples the second bus **31** to the connecting node **138**.

[0204] The inductor **133** permits to limit current peaks in the string **130** and the phases of the grid **G** during commutations of the input switching units **77**.

[0205] In a variant, the string **130** does not comprise the inductor **133**, the ends **136**, **135** of the first and second portions **131**, **132** and the input **40a** of the input module **40** being connected to the connecting node **138**, the first portion **131** or the second portion **132** comprising an inductor to limit the current peaks in the string **130** and the string **130** comprises another inductor connecting the input **40a** of the input module **40** to the connecting node **130** to limit the current peaks in the phase of the grid **G**.

[0206] In another variant, the string **130** does not comprise the inductor **133**, the ends **136**, **135** of the first and second portions **131**, **132** and the input **40a** of the input module **40** are connected to the connecting node **138**, each of the first portion **131** and the second portion **132** comprising an inductor to limit the current peaks in the string **130** and the string **130** may comprise another inductor connecting the input **40a** of the input module **40** to the connecting node **138** to limit the current peaks in the phase of the grid **G**.

[0207] As the output modules **49**, **50**, **51** are identical, FIG. 7 illustrates schematically a first example of the output module **49** comprising a first example of the HMMC.

[0208] The first example of the output module **49** comprises a leg **140** comprising a first string **141** and a second string **142**.

[0209] The first string **141** comprises a first branch **143** and a second branch **144**.

[0210] Each branch **143**, **144** comprises a first end **145**, **146** and a second end **147**, **148**.

[0211] The first end **145** of the first branch **143** is connected to the first input **49a** of the output module **49** and the second end **148** of the second branch **144** is connected to the second input **49b** of the output module **49**.

[0212] Each branch **143**, **144** comprises a first set of output switches **149**, **150**, **151**, **152** connected in series between the first and the second ends **145**, **146**, **147**, **148** of the branches **143**, **144**.

[0213] The first branch **143** further comprises a first connecting node **153** situated between two output switches **149**, **150** of the first set of output switches so that the number of output switches between the first connecting node **153** and

the first end **145** of the branch is equal to the number of output switches between the first connecting node **153** and the second end **147**.

[0214] The second branch **144** further comprises a second connecting node **154** situated between two output switches **151**, **152** of the second set of output switches so that the number of output switches between the second connecting node **154** and the first end **146** of the branch is equal to the number of output switches between the second connecting node **154** and the second end **148**.

[0215] Each branch **143**, **144** may comprise more than two output switches, the two branches comprising a same number of output switches.

[0216] Each output switch **149**, **150**, **151**, **152** comprises a first end **149a**, **150a**, **151a**, **152a** and a second end **149b**, **150b**, **151b**, **152b**.

[0217] The output switches **149**, **150** of the first branch **143** are connected in series so that the first end **150a** of the output switch **150** is connected to the first end **145** of the first branch **143**, the second end **149b** and the first end **150a** of the output switches **149**, **150** are connected to the first connecting node **153**, and the second end **150b** of the switch **150** is connected to the second end **147** of the first branch **143**.

[0218] The output switches **151**, **152** of the second branch **144** are connected in series so that the first end **151a** of the output switch **151** is connected to the first end **146** of the second branch **144**, the second end **151b** and the first end **152a** of the output switches **151**, **152** are connected to the second connecting node **154**, and the second end **152b** of the output switch **152** is connected to the second end **148** of the second branch **144**.

[0219] The first string **141** further comprises a third connecting node **155** connected to the second end **147** of the first branch **143**, the first end **146** of the second branch **144**, and the third input **49c** of the output module **49**.

[0220] The first set of output switches **149**, **150** operatively couple the first bus **30** to the third connecting node **155**, and the second set of output switches **151**, **152** operatively couple the second bus **31** to the third connecting node **155**.

[0221] The second string **142** comprises a first portion **156**, a second portion **157**, and a fourth connecting node **158**.

[0222] The second string **142** may further comprise an inductor **159a**.

[0223] Each portion **156**, **157** comprises a first end **159**, **160**, a second end **161**, **162**.

[0224] The first portion **156** comprises a first set of output switching units **163** in series between the first and second ends **159**, **161** of the portion **156**, **157**.

[0225] The second portion **157** comprises a second set of output switching units **163** in series between the first and second ends **160**, **162** of the portion **156**, **157**.

[0226] Each set of output switching units **163** comprises a same number of output switching units **163**.

[0227] Each output switching unit **163** includes a plurality of controllable switches and energy storage means, the controllable switches being intended to bridge the energy storage means or to connect the energy storage means in each output switching unit **163**.

[0228] The output switching units **163** of each set are connected in series so that the first connection **164** of an

output switching unit 163 is connected to a second connection 165 of an adjacent output switching unit 163.

[0229] The first remaining connection of the output switching unit 163 of the first set is connected to the first end 159 and the second remaining connection 165 of the output switching unit 163 of the first set is connected to the second end 161.

[0230] The first end 159 of the first portion 156 is connected to the first connecting node 153 and the second end 161 is connected to a first end of the inductor 159a.

[0231] The first remaining connection of the output switching unit 163 of the second set is connected to the first end 160 and the second remaining connection of the output switching unit 163 of the second set is connected to the second end 162.

[0232] The first set of output switching units 163 operatively couples the first connecting node 153 to the fourth connecting node 158 and the second set of output switching units 163 operatively couples the second connecting node 154 to the fourth connecting node 158.

[0233] The first end 160 of the second portion 157 is connected to a second end of the inductor 159a and the second end 162 is connected to the second connecting node 154.

[0234] The fourth connecting node 158 connects the output 49d of the output module 49 to a mid-connecting node of the inductor 159a.

[0235] The inductor 159a permits to limit current peaks in the second string 142 and the phases of the arc furnace 2 during commutations of the output switching units 163.

[0236] In a variant, the second string 142 does not comprise the inductor 159a, the ends 161, 160 of the first and second portions 156, 157 and the output 49d of the output module 49 being connected to the fourth connecting node 158, the first portion 156 or the second portion 157 comprising an inductor to limit the current peaks in the second string 142 and the second string 142 comprises another inductor connecting the output 49d of the output module 49 to the fourth connecting node 159 to limit the current peaks in the phase of the arc furnace 2.

[0237] In another variant, the second string 142 does not comprise the inductor 159a, the ends 161, 160 of the first and second portions 156, 157 and the output 49d of the output module 49 are connected to the fourth connecting node 158, each of the first portion 156 and the second portion 157 comprising an inductor to limit the current peaks in the second string 142 and the second string 142 may comprise another inductor connecting the output 49d of the output module 49 to the fourth connecting node 158 to limit the current peaks in the phase of the arc furnace 2.

[0238] FIG. 8 illustrates schematically a second embodiment of the output module 49 comprising a second embodiment of the hybrid modular multilevel converter HMMC.

[0239] The second example of the output module 49 comprises a leg 170 comprising a first string 171 and a second string 172.

[0240] The first string 171 comprises a first branch 173 and a second branch 174.

[0241] Each branch 173, 174 comprises a first end 175, 176 and a second end 177, 178.

[0242] The first end 175 of the first branch 173 is connected to the first input 49a of the output module 49 and the second end 178 of the second branch 174 is connected to the second input 49b of the output module 49.

[0243] The first branch 173 comprises a first output switch 179, a first connecting node 180, a first set 181 of output switching units 163 connected in series, and a first inductor 182.

[0244] A first end 179a of the first output switch 179 is connected to the first end 175 of the first branch 173 and a second end 179b of the first output switch 179 is connected to the first connecting node 180.

[0245] The output switching units 163 of the first set 181 are connected in series so that the first connection 164 of an output switching unit 163 is connected to a second connection 165 of an adjacent output switching unit 163.

[0246] The first remaining connection of the output switching unit 163 of the first set 181 is connected to the first connecting node 180 and the second remaining connection of the output switching unit 163 of the first set 181 is connected to a first end of the first inductor 182.

[0247] The second end of the first inductor 182 is connected to the second end 177 of the first branch 173.

[0248] The second branch 174 comprises a second output switch 183, a second connecting node 184, a second set 185 of output switching units 163 connected in series, and a second inductor 186.

[0249] A second end 183b of the second output switch 183 is connected to the second end 178 of the second branch 174 and a first end 183a of the second output switch 183 is connected to the second connecting node 184.

[0250] The output switching units 163 of the second set 185 are connected in series so that the first connection 164 of an output switching unit 163 is connected to a second connection 165 of an adjacent output switching unit 163.

[0251] The second remaining connection 165 of the output switching unit 163 of the second set 185 is connected to the second connecting node 184 and the first remaining connection 164 of the output switching unit 163 of the second set 185 is connected to a first end of the second inductor 186.

[0252] The second end of the second inductor 186 is connected to the first end 176 of the second branch 174.

[0253] The first and second sets 181, 185 of output switching units 163 have a same number of output switching units 163.

[0254] The first string 171 further comprises a third connecting node 187 connected to the first end 176 of the second branch 174, the second end 177 of the first branch 173 and the third input 49c of the output module 49.

[0255] The first output switch 179 is operatively coupling the first bus 30 to the first connecting node 180 and the first set 181 of output switching units 163 is operatively coupling the first connecting node 180 to the third connecting node 187.

[0256] The second output switch 183 is operatively coupling the second bus 31 to the second connecting node 184 and the second set 185 of output switching units 163 is operatively coupling the second connecting node 184 to the third connecting node 187.

[0257] Each branch 173, 174 may comprise more than one output switch, the output switches being connected in series between a connecting node and one end of the branch.

[0258] The first and second inductors 182, 186 limit current peaks in the first string 171 and the phases of the arc furnace 2 during commutations of the output switching units 163.

[0259] The second string 172 comprises a first portion 186 and a second portion 187.

[0260] Each portion 186, 187 comprises a first end 188, 189 and a second end 190, 191.

[0261] The first portion 186 comprises a third output switch 192 comprising a first end 192a connected to the first end 188 of the portion 186, 187 and a second end 192b connected to the second end 190 of the portion 186, 187.

[0262] The second portion 187 comprises a fourth output switch 193 comprising a first end 193a connected to the first end 189 of the portion 186, 187 and a second end 193b connected to the second end 191 of the portion 186, 187.

[0263] In another embodiment, each portion 186, 187 may comprise more than one output switch connected in series between both ends of the portion 186, 187, the number of output switches of the first and second portions being identical.

[0264] The second string 172 further comprises a fourth connecting node 194 connected to the second end 190 of the first portion 186, the first end 189 of the second portion 187, and the output 49d of the output module 49.

[0265] The first end 188 of the first portion 186 is connected to the first connecting node 180 and the second end 191 of the second portion 187 is connected to the second connecting node 184.

[0266] The third output switch 192 operatively couples the first connecting node 180 to the fourth connecting node 194 and the fourth output switch 193 operatively couples the second connecting node 184 to the fourth connecting node 194.

[0267] FIG. 9 illustrates schematically a third embodiment of the output module 49 comprising a third embodiment of the hybrid modular multilevel converter HMMC.

[0268] The third embodiment of the output module 49 comprises a leg 195 comprising a first string 196 and the second string 172 according to the second example of the hybrid modular multilevel converter illustrated in FIG. 8.

[0269] The first string 196 comprises a first branch 197 and a second branch 198.

[0270] Each branch 197, 198 comprises a first end 199, 200 and a second end 201, 202.

[0271] The first end 199 of the first branch 197 is connected to the first input 49a of the output module 49 and the second end 202 of the second branch 198 is connected to the second input 49b of the output module 49.

[0272] The first branch 197 comprises a first output switch 203, a first connecting node 204, a first inductor 205, and a first set 206 of output switching units 163 connected in series.

[0273] A second end 203b of the first output switch 203 is connected to the second end 201 of the first branch 197 and a first end 203 of the first output switch 203 is connected to the first connecting node 204.

[0274] The output switching units 163 of the first set 206 are connected in series so that the first connection 164 of an output switching unit 163 is connected to a second connection 165 of an adjacent output switching unit 163.

[0275] The first remaining connection 164 of the output switching unit 163 of the first set 206 is connected to the first end 199 of the first branch 197 and the second remaining connection 165 of the output switching unit 163 of the first set 206 is connected to a first end of the first inductor 205.

[0276] The second end of the first inductor 205 is connected to the first connecting node 204.

[0277] The second branch 198 comprises a second output switch 207, a second connecting node 208, a second inductor 209, and a second set 210 of input switching units 163 connected in series.

[0278] A first end 207a of the second output switch 207 is connected to the first end 200 of the second branch 198 and a second end 207b of the second output switch 207 is connected to the second connecting node 208.

[0279] The output switching units 163 of the second set 210 are connected in series so that the first connection 164 of an output switching unit 163 is connected to a second connection 165 of an adjacent switching unit 163.

[0280] The first remaining connection 164 of the output switching unit of the second set 210 is connected to a first end of the second inductor 209.

[0281] A second end of the second inductor 209 is connected to the second connecting node 208.

[0282] The second remaining connection 165 of the output switching unit 163 of the second set 210 is connected to the second end 202 of the second branch 198.

[0283] The first and second sets 206, 210 of output switching units 163 have a same number of output switching units 163.

[0284] The first string 196 further comprises a third connecting node 211 connected to the first end 200 of the second branch 198, the second end 201 of the first branch 197 and the third input 49c of the output module 49.

[0285] The first set 206 of output switching units 163 operatively couples the first bus 30 to the first connecting node 204 and the first output switch 203 operatively coupling the first connecting node 204 to the third connecting node 211.

[0286] The second set 210 of output switching units 163 operatively couples the second bus 31 to the second connecting node 208 and the second input switch 207 operatively couples the second connecting node 208 to the third connecting node 211.

[0287] Each branch 197, 198 may comprise more than one output switch, the output switches being connected in series between a connecting node and one end of the branch 197, 198.

[0288] The first and second inductors 205, 209 limit current peaks in the first string 196 and the phases of the arc furnace 2 during commutations of the output switching units 163.

[0289] The first end 188 of the first portion 186 of the second string 172 is connected to the first connecting node 204 and the second end 191 of the second portion 187 of the second string 172 is connected to the second connecting node 208.

[0290] FIG. 10 illustrates schematically a fourth embodiment of the output module 49 comprising an embodiment of a modular multi-level converter MMC.

[0291] The output module 49 comprises a string 220 including a first portion 221 and a second portion 222.

[0292] The output module 49 comprises an inductor 223.

[0293] Each portion 221, 222 comprises a first end 224, 225, a second end 226, 227.

[0294] The first end 224 of the first portion 221 is connected to the first input 49a of the output module 49 and the second end 227 of the second portion 222 is connected to the second input 49b of the output module 49.

[0295] The input 49c of the output module 49 is not connected.

[0296] The first portion 221 comprises a first set of output switching units 163 in series between the first and second ends 224, 226 of the portion 221, 222.

[0297] The second portion 222 comprises a second set of output switching units 163 in series between the first and second ends 225, 227 of the portion 221, 222.

[0298] Each set of output switching units 163 comprises a same number of output switching units 163.

[0299] The output switching units 163 of each set are connected in series so that the first connection 164 of an output switching unit 163 is connected to a second connection 165 of an adjacent output switching unit 163.

[0300] The first remaining connection 164 of the output switching unit 163 of the first set is connected to the first end 224 and the second remaining connection 165 of the output switching unit 163 of the first set is connected to the second end 226.

[0301] The first remaining connection 164 of the output switching unit 163 of the second set 222 is connected to the first end 225 and the second remaining connection 165 of the output switching unit 163 of the second set is connected to the second end 227.

[0302] The string 220 further comprises a connecting node 228.

[0303] The second end 226 of the first portion 221 is connected to a first end of the inductor 223 and the first end 225 of the second portion 222 is connected to the second end of the inductor 223.

[0304] The connecting node 228 connects the output 49d of the output module 49 to a mid-connecting node of the inductor 223.

[0305] The first set of output switching units 163 operatively couples the first bus 30 to the connecting node 228 and the second set of output switching units 163 operatively couples the second bus 31 to the connecting node 228.

[0306] The inductor 223 permits to limit current peaks in the string 220 and the phases of the grid G during commutations of the output switching units 163.

[0307] In another embodiment, the string 220 does not comprise the inductor 223, the ends 226, 225 of the first and second portions 221, 222 and the output 49d of the output module 49 being connected to the connecting node 228, the first portion 221 or the second portion 222 comprising an inductor to limit the current peaks in the string 220 and the string 220 comprises another inductor connecting the output 49d of the output module 49 to the connecting node 228 to limit the current peaks in the phase of the arc furnace 2.

[0308] In another embodiment, the string 220 does not comprise the inductor 223, the ends 226, 225 of the first and second portions 221, 222 and the output 49d of the output module 49 are connected to the connecting node 228, each of the first portion 221 and the second portion 222 comprising an inductor to limit the current peaks in the string 220 and the string 220 may comprise another inductor connecting the output 49d of the output module 49 to the connecting node 228 to limit the current peaks in the phase of the arc furnace 2.

[0309] FIG. 11 illustrates schematically a first embodiment of the input switches 61, 62, 63, 64, 89, 93, 104, 105, 118, 122 and the output switches 149, 150, 151, 152, 179, 183, 192, 193, 203, 207.

[0310] Each switch of the input and output switches 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 comprises a diode 230 extend-

ing between the first and the second ends of the switch 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 so that the anode of the diode 230 is connected to the second end 61b, 62b, 63b, 64b, 89b, 93b, 104b, 105b, 118b, 122b, 149b, 150b, 151b, 152b, 179b, 183b, 192b, 193b, 203b, 207b of the switch and the cathode is connected to the first end 61a, 62a, 63a, 64a, 89a, 93a, 104a, 105a, 118a, 122a, 149a, 150a, 151a, 152a, 179a, 183a, 192a, 193a, 203a, 207a of the switch.

[0311] As the diode 230 does need to be controlled by the control circuit 11, the first example of the switch comprising the diode 230 simplify the implementation of the input switches in the input modules 40, 41, 42 and in the implementation of the output switches in the output modules 49, 50, 51.

[0312] Each switch 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 may comprise a plurality of diodes 230 in series so that the anode of one diode is connected to the cathode of another diode.

[0313] When the input switches 61, 62, 63, 64, 89, 93, 104, 105, 118, 122 according to the first embodiment comprising the diode 230 are implemented in input modules comprising an HMMC as illustrated in FIGS. 3 to 5 according to the first, second and third embodiments of the input modules or when the output switches 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 according to the first embodiment comprising the diode 230 are implemented in output modules comprising an HMMC as illustrated in FIGS. 6 to 8 according to the first, second and third embodiments of the output modules, the diode 230 is designed to support a voltage at least equal to VDC/2.

[0314] Each input or output switch 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 may comprise a plurality of diodes 230 in series so that the anode of one diode is connected to the cathode of another diode.

[0315] FIG. 12 illustrates schematically a first embodiment of the input switches 61, 62, 63, 64, 89, 93, 104, 105, 118, 122 and the output switches 149, 150, 151, 152, 179, 183, 192, 193, 203, 207.

[0316] Each switch of the input and output switches 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 comprises the diode 230 extending between the first and the second ends of the switch 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 so that the anode of the diode 230 is connected to the second end 61b, 62b, 63b, 64b, 89b, 93b, 104b, 105b, 118b, 122b, 149b, 150b, 151b, 152b, 179b, 183b, 192b, 193b, 203b, 207b of the switch and the cathode is connected to the first end 61a, 62a, 63a, 64a, 89a, 93a, 104a, 105a, 118a, 122a, 149a, 150a, 151a, 152a, 179a, 183a, 192a, 193a, 203a, 207a of the switch.

[0317] Each switch 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 further comprises a gate turn off thyristor 231.

[0318] The gate of the gate turn off thyristor 231 is connected to the control circuit 11, the anode of the gate turn off thyristor 231 is connected to the cathode of the diode 230, and the cathode of the gate turn off thyristor 231 is connected to the anode of the diode 230.

[0319] According to an embodiment, each switch 61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207 may comprise a plurality of gate of

the gate turn off thyristors **231** in series so that the anode of one thyristor is connected to the cathode of another thyristor.

[0320] When the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122** according to the second embodiment comprising the diode **230** and the gate turn off thyristor **230** are implemented in input modules comprising an HMMC as illustrated in FIGS. **3 to 5** according to the first, second and third embodiments of the input modules or when the output switches **149, 150, 151, 152, 179, 183, 192, 193, 203, 207** according to the second embodiment are implemented in output modules comprising an HMMC as illustrated in FIGS. **6 to 8** according to the first, second and third embodiments of the output modules, the diode **230** and the gate turn off thyristors **231** are designed to support a voltage at least equal to $VDC/2$.

[0321] Each input or output switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** may comprise a plurality of gate turn off thyristors **231** in parallel with the diode **230** in series so that the anode of one diode is connected to the cathode of another diode.

[0322] FIG. **13** illustrates schematically a third embodiment of the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122** and the output switches **149, 150, 151, 152, 179, 183, 192, 193, 203, 207**.

[0323] Each switch of the input and output switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** comprises the diode **230** extending between the first and the second ends of the switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** so that the anode of the diode **230** is connected to the second end **61b, 62b, 63b, 64b, 89b, 93b, 104b, 105b, 118b, 122b, 149b, 150b, 151b, 152b, 179b, 183b, 192b, 193b, 203b, 207b** of the switch and the cathode is connected to the first end **61a, 62a, 63a, 64a, 89a, 93a, 104a, 105a, 118a, 122a, 149a, 150a, 151a, 152a, 179a, 183a, 192a, 193a, 203a, 207a** of the switch.

[0324] Each switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** further comprises a field effect transistor **232**.

[0325] The field effect transistor **232** may be an insulated gate bipolar transistor for example using silicon-base semiconductors such as insulated gate bipolar transistor IGBT or injection-enhanced gate transistor IEGT, or using silicon carbide semiconductors such as metal-oxide-semiconductor field-effect transistor MOSFET.

[0326] The gate of the transistor **232** is connected to the control circuit **11**, the drain of the transistor **232** is connected to the cathode of the diode **230** and the source of the transistor **232** is connected to the anode of the diode **230**.

[0327] Each switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** may comprise a plurality of transistors **232** and diodes **230** in parallel in series so that the drain of one transistor is connected to the source of another transistor.

[0328] When the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122** according to the third embodiment are implemented in input modules comprising an HMMC as illustrated in FIGS. **3 to 5** according to the first, second and third embodiments of the input modules or when the output switches according to the third embodiment of the are implemented in output modules comprising an HMMC as illustrated in FIGS. **6 to 8** according to the first, second and third embodiments of the output modules, the diode **230** is designed to support a voltage at least equal to $VDC/2$.

[0329] The inputs modules **40, 41, 42** may comprise one embodiment of the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122** illustrated in FIGS. **11, 12, 13**, and the output modules **49, 50, 51** may comprise the same or another embodiment of the output switches **149, 150, 151, 152, 179, 183, 192, 193, 203, 207** illustrated in FIGS. **11, 12, 13**.

[0330] FIG. **14** illustrates schematically a first embodiment of the input switching unit **77** and the output switching unit **163**.

[0331] Each switching unit of the input switching unit **77** and the output switching unit **163** comprises two controllable switches **233, 234** and an energy storage means **235** comprising a first end **235a** and a second end **235b**.

[0332] The energy storage means **235** comprises for example a capacitor.

[0333] Each controllable switch **233, 234** comprises a first end **233a, 234a**, a second end **233b, 234b**, and a control input **233c, 234c** connected to the control circuit **11**.

[0334] The first end **233a** of a first controllable switch **233** is connected to the first end **235a** of the energy storage means **235**.

[0335] The second end **233b** of the first controllable switch **233** and the first end **234a** of the second controllable switch **234** are connected to the first end **78, 164** of the input switching unit **77** or output switching unit **163**.

[0336] The second end **234b** of the second controllable switch **234** is connected to the second end **235b** of the energy storage means **235** and to the second end **79, 165** of the input switching unit **77** or output switching unit **163**.

[0337] As the controllable switches **233, 234** are identical, only the first switch **233** is detailed.

[0338] The first switch **233** comprises a transistor **236** and a diode **237**.

[0339] The transistor **236** may be a field effect transistor.

[0340] The field effect transistor **236** may be an insulated gate bipolar transistor for example using silicon-base semiconductors such as insulated gate bipolar transistor IGBT or injection-enhanced gate transistor IEGT, or using silicon carbide semiconductors such as metal-oxide-semiconductor field-effect transistor MOSFET.

[0341] The gate of the transistor **236** is connected to the control input **233c**, the drain of the transistor **236** and the cathode of the diode **237** are connected to the first end **233a**, and the source of the transistor **236** and the anode of the diode **237** are connected to the second end **233b**.

[0342] FIG. **15** illustrates schematically a second embodiment of the input switching unit **77** and the output switching unit **163**.

[0343] Each switching unit of the input switching unit **77** and the output switching unit **163** comprises four controllable switches **238, 239, 240, 241** identical to the controllable switches **233, 234** of the first embodiment illustrated in FIG. **14**.

[0344] Each controllable switches **238, 239, 240, 241** comprises a first end **238a, 239a, 240a, 241a**, a second end **238b, 239b, 240b, 241b**, and a control input **238c, 239c, 240c, 241c** connected to the control circuit **11**.

[0345] Each switching unit **77, 163** further comprises the energy storage means **235**.

[0346] The first end **238a** of a first controllable switch **238** and the first end **239a** of a second controllable switch **239** are connected to the first end **235a** of the energy storage means **235**.

[0347] The second end **240b** of a third controllable switch **240** and the second end **241b** of the fourth controllable switch **241** are connected to the second end **235b** of the energy storage means **235**.

[0348] The second end **238b** of the first controllable switch **238** and the first end **240a** of the third controllable switch **240** are connected to the first end **78**, **164** of the input switching unit **77** or output switching unit **163**.

[0349] The second end **239b** of the second controllable switch **239** and the first end **241a** of the fourth controllable switch **241** are connected to the second end **79**, **165** of the input switching unit **77** or output switching unit **163**.

[0350] The input switching units **77** and the output switching units **163** according to the first embodiment illustrated in FIG. **14** have two states, each state delivering a different voltage value on the ends of the input switching units **77** and the output switching units **163**.

[0351] For example, in a first state, the input and output switching units **77**, **163** deliver a positive compensating voltage V_{cc} , and in a second state a nil voltage.

[0352] The input switching units **77** and the output switching units **163** according to the second embodiment illustrated in FIG. **15** have three states, each state delivering a different voltage value on the ends of the input switching units **77** and the output switching units **163**.

[0353] For example, in a first state, the input and output switching units **77**, **163** deliver a positive voltage V_{cc} , in a second state a nil voltage, and in a third state a negative compensating voltage $-V_{cc}$.

[0354] The embodiment of the input switching units **77** of inputs modules **40**, **41**, **42** and the embodiment of the output switching units **163** of the output modules **49**, **50**, **51** may be identical or different.

[0355] FIG. **16** illustrates a second embodiment of the arc furnace facility.

[0356] The arc furnace facility differs from the arc furnace facility illustrated in FIG. **1** in that the arc furnace facility comprises a direct voltage arc furnace **250**.

[0357] According to an embodiment, the direct voltage arc furnace **250** comprises the set of electrodes comprising only one electrode **250a**.

[0358] To supply the direct voltage arc furnace **250**, the power supply system **1** further comprises a rectifier **251**.

[0359] The rectifier **251** comprises a plurality of inputs **252**, **253**, **254**, being connected the phase **25**, **26**, **27** of the secondary circuit of the transformer **10**.

[0360] The rectifier **251** further comprises a first output **255** connected to an enclosure **250b** of the arc furnace **250** and a second output **256** connected to the electrode **250a** of the arc furnace **250**.

[0361] The rectifier **251** comprises the same number of identical legs **257**, **258**, **259** as the number of phases of the secondary circuit of the transformer **10**, each leg **257**, **258**, **259** being connected to a different phase.

[0362] Each leg **257**, **258**, **259** comprises a first connection **257a**, **258a**, **259a**, a second connection **257b**, **258b**, **259b**, and a third connection **257c**, **258c**, **259c**.

[0363] The first connection **257a**, **258a**, **259a** of each leg **257**, **258**, **259** is connected to the first output **255** of the rectifier **251**, the second connection **257b**, **258b**, **259b** of each leg **257**, **258**, **259** is connected to the second output **256** of the rectifier **251**, and the third connection **257c**, **258c**, **259c** is connected to a different input **252**, **253**, **254** of the rectifier **251**.

[0364] Since the legs **257**, **258**, **259** are identical, for illustration only a first leg **257** is described herein.

[0365] The first leg **257** comprises two identical power diodes **260**, **261** connected in series.

[0366] The cathode of a first power diode **260** is connected to the first connection **257a**, the anode of the first power diode **260** and the cathode of the second power diode **261** are connected to the third connection **257c**, and the anode of the second power diode **261** is connected to the second connection **257b**.

[0367] The rectifier **251** rectifies the voltage delivered by each phase of the secondary circuit to supply the electrode **250a** of the direct voltage arc furnace **250**.

[0368] In a variant, the rectifier **251** may comprise an active rectifier made from controlled semiconductor, for example transistors and/or gate turn off thyristors.

[0369] The implementation of a HMMC according to the examples described above in input and output modules permits to reduce the volume, the size and the weight of the modules, and to increase power density and efficiency of the modules.

[0370] A first example of a method implementing the power supply system **1** is presented.

[0371] The control circuit **11** opens the first disconnecting means **13** and the first isolation switch **15A** and closes the second disconnecting means **14**, the second isolation switch **15B** and if applicable the third isolation switch **15C**.

[0372] As first disconnecting means **13** and first isolation switch **15A** are open, no current flows through the first disconnecting means **13**, and first isolation switch **15A**.

[0373] As the second disconnecting means **14** are closed, currents flow from the outputs **19**, **20**, **21** of the power converter **9** to the primary circuit of the transformer **10**.

[0374] The control circuit **11** controls the input modules **40**, **41**, **42** and the outputs modules **49**, **50**, **51** to supply the set of electrodes and to stabilize the current and the voltage delivered by the grid **G** to reduce reactions in the grid **G** generated by the arc furnace **2**, **250**.

[0375] The control circuit **11** controls independently each input switching unit **77** and each output switching unit **163**, and if applicable each input switch and each output switch to supply the set of electrodes and to stabilize the current and the voltage delivered by the grid **G** to reduce reactions in the grid **G**.

[0376] In a second embodiment of the method implementing the power supply system **1**, the control circuit **11** closes the first disconnecting means **13**, the second isolation switch **15B** so that currents flow through the second isolation switch **15B** and if applicable the third isolation switch **15C** so that currents flow through the third isolation switch **15C**, and opens the second disconnecting means **14** and the first isolation switch **15A** so that no current flows from the outputs of the power converter **9** to the primary circuit, the primary circuit being supplied directly by the grid **G**.

[0377] The output switching unit **163** are idle and if applicable the output switches are open.

[0378] The control circuit **11** controls independently each input switching unit **77** of the input modules to stabilize the current and the voltage delivered by the grid **G** to reduce reactions in the grid **G**.

[0379] The input modules of the power converter **9** acts as a static compensator STATCOM to reduce the reactions on the grid **G**.

[0380] When the power converter 9 comprises the input modules 40, 41, 42 according to the first example illustrated in FIG. 3, in this second example of the method, the control circuit 11 further closes the input switch 61 of the first set of input switches between the first end 57 of the first branch 55 and the first connecting node 65 of the first branch 55, and the input switch 64 of the second set of input switches between the second end 60 of the second branch 56 and the second connecting node 66 of the second branch 56, and the control circuit 11 further opens the input switch 62 of the first set of input switches between the first connecting node 65 of the first branch 55 and the third connecting node 67 of the first string 53 and, the input switch 63 of the second set of input switches between the second connecting node 66 of the second branch 56 and the third connecting node 67 of the first string 53.

[0381] In another embodiment, the control circuit 11 further opens the input switch 61 of the first set of input switches between the first end 57 of the first branch 55 and the first connecting node 65 of the first branch 55, and the input switch 64 of the second set of input switches between the second end 60 of the second branch 56 and the second connecting node 66 of the second branch 56, and the control circuit 11 further closes the input switch 62 of the first set of input switches between the first connecting node 65 of the first branch 55 and the third connecting node 67 of the first string 53 and, the input switch 63 of the second set of input switches between the second connecting node 66 of the second branch 56 and the third connecting node 67 of the first string 53.

[0382] When the power converter 9 comprises the input modules 40, 41, 42 according to the second example illustrated in FIG. 4, in this second embodiment of the method, the control circuit 11 further opens the first input switch 89 and the second input switch 93, and further closes the third input switch 104 and the fourth input switch 105.

[0383] When the power converter 9 comprises the input modules 40, 41, 42 according to the third example illustrated in FIG. 5, in this second embodiment of the method, the control circuit 11 further opens the first input switch 118 and the second input switch 122, and further closes the third input switch 104 and the fourth input switch 105.

[0384] In a third embodiment of the method implementing the power supply system 1, the control circuit 11 opens the first disconnecting means 13 so that currents flow from the grid G to the inputs 16, 17, 18 of the power converter 9 and so that no current flows directly from the grid G to the second disconnecting means 14, opens the second isolation switch 15B and if applicable the third isolation switch 15C, and closes the second disconnecting means 14 and the first isolation switch 15A, so that electrode(s) of the arc furnace 2, 250 is directly supplied by the power converter 9, the transformer 10 being bypassed to simplify the control of the power converter 9 to supply the arc furnace 2, 250, and to improve the reliability and the efficiency of the power supply system 1.

[0385] The first and second isolation switches 15A, 15B permit to bypass and isolate the transformer 10.

[0386] The third isolation switch 15C permits to bypass the transformer 10.

1. A power supply system for arc furnace, the power supply system comprising:

a power converter comprising an input intended to be connected to a polyphase supply grid and an output, and

a polyphase transformer comprising a primary circuit and a secondary circuit,

a first disconnecting means, and

a second disconnecting means,

the power supply system further comprises a first isolation switch and a second isolation switch, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch, and

a control circuit configured to:

control the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and

control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed.

2. The power supply system according to claim 1, wherein the power converter comprises an input device, a link circuit comprising a first bus and a second bus, and an output device, wherein:

the input device comprises a plurality of input modules, each input module being connected to the first bus and the second bus and being intended to be connected to a different phase of the supply grid to supply the link circuit with a continuous voltage from the phase of the supply grid,

the output device comprises a plurality of output modules, each output module being connected to the first bus and second bus and to the primary circuit to supply the transformer with an alternative voltage from the circuit link.

3. The power supply system according to claim 2, wherein each input module comprises:

a leg including:

a first string comprising:

a first set of input switches connected in series, a second set of input switches connected in series, a first connecting node, a second connecting node, and a third connecting node,

the first set of input switches operatively coupling the first bus to the third connecting node, and the first connecting node being disposed between two input switches of the first set of input switches,

the second set of input switches operatively coupling the second bus to the third connecting node, and

the second connecting node being disposed between two input switches of the second set of input switches,

a second string comprising:

- a first set of input switching units in series, a second set of input switching units in series, and a fourth connecting node,
- the first set of input switching units operatively coupling the first connecting node to the fourth connecting node,
- the second set of input switching units operatively coupling the second connecting node to the fourth connecting node, and
- the fourth connecting node being intended to be connected to the phase of the supply grid,

the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

4. The power supply system according to claim 2, wherein each input module comprises:

a leg including:

a first string comprising:

- a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series,

- the first input switch operatively coupling the first bus to the first connecting node, and the first set of input switching units operatively coupling the first connecting node to the third connecting node,

- the second input switch operatively coupling the second bus to the second connecting node, and the second set of input switching units operatively coupling the second connecting node to the third connecting node,

a second string comprising:

- a third input switch, a fourth input switch, and a fourth connecting node,

- the third input switch operatively coupling the first connecting node to the fourth connecting node, and the fourth input switch operatively coupling the second connecting node to the fourth connecting node,

- the fourth connecting node (106) being intended to be connected to the phase of the supply grid,

the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes (97) of the input modules being connected together.

5. The power supply system according to claim 2, wherein each input module comprises:

a leg including:

a first string comprising:

- a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series,

- the first set of input switching units operatively coupling the first bus to the first connecting node, and the first input switch operatively coupling the first connecting node to the third connecting node,

- the second set of input switching units operatively coupling the second bus to the second connecting node, and the second input switch operatively coupling the second connecting node to the third connecting node,

a second string comprising:

- a third input switch, a fourth input switch, and a fourth connecting node,

- the third input switch operatively coupling the first connecting node to the fourth connecting node, and the fourth input switch operatively coupling the second connecting node to the fourth connecting node,

- the fourth connecting node being intended to be connected to the phase of the supply grid,

the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

6. The power supply system according to claim 2, wherein each input module comprises a string comprising:

- a connecting node being intended to be connected to the phase of the supply grid, a first set of input switching units connected in series, and a second set of input switching units connected in series,

- the first set of input switching units operatively coupling the first bus to the connecting node, and the second set of input switching units operatively coupling the second bus to the connecting node, and

- the control circuit being configured to control the input switching units, and the output modules.

7. The power supply system according to claim 6, wherein each output module comprises:

a leg including:

a first string comprising:

- a first set of output switches connected in series, a second set of output switches connected in series, a first connecting node, a second connecting node, and a third connecting node,

- the first set of output switches operatively coupling the first bus to the third connecting node, and the first connecting node being disposed between two output switches of the first set of output switches,

- the second set of output switches operatively coupling the second bus to the third connecting node, and the second connecting node being disposed between two output switches of the second set of output switches,

a second string comprising:

- a first set of output switching units in series, a second set of output switching units in series, and a fourth connecting node,

- the first set of output switching units operatively coupling the first connecting node to the third connecting node,

- the second set of output switching units operatively coupling the second connecting node to the fourth connecting node, and

- the fourth connecting node being intended to be connected to a phase of the arc furnace,

the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

8. The power supply system according to claim 6, wherein each output module comprises:

a leg including:

a first string comprising:

a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series,

the first output switch operatively coupling the first bus to the first connecting node, and the first set of output switching units operatively coupling the first connecting node to the third connecting node, the second output switch operatively coupling the second bus to the second connecting node, and the second set of output switching units operatively coupling the second connecting node to the third connecting node,

a second string comprising:

a third output switch, a fourth output switch, and a fourth connecting node,

the third output switch operatively coupling the first connecting node to the fourth connecting node, and the fourth output switch operatively coupling the second connecting node to the fourth connecting node,

the fourth connecting node being intended to be connected to a phase of the arc furnace,

the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

9. The power supply system according to claim 6, wherein each output module comprises:

a leg including:

a first string comprising:

a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series,

the first set of output switching units operatively coupling the first bus to the first connecting node, and the first output switch operatively coupling the first connecting node to the third connecting node,

the second set of output switching units operatively coupling the second bus to the second connecting node, and the second output switch operatively coupling the second connecting node to the third connecting node,

a second string comprising:

a third output switch, a fourth output switch, and a fourth connecting node,

the third output switch operatively coupling the first connecting node to the fourth connecting node, and the fourth output switch operatively coupling the second connecting node to the fourth connecting node,

the fourth connecting node being intended to be connected to a phase of the arc furnace,

the control circuit being configured to control the output switches, the output switching units, and the output

modules, the third connecting nodes of the output modules being connected together.

10. The power supply system according to claim 6, wherein each output module comprises a string comprising:

a connecting node being intended to be connected to a phase of the arc furnace, a first set of output switching units connected in series, and a second set of output switching units connected in series,

the first set of output switching units operatively coupling the first bus to the connecting node, and the second set of output switching units operatively coupling the second bus to the connecting node,

the control circuit being configured to control the output switching units and the output modules.

11. The power supply system according to claim 10, further comprising a third isolation switch, the third isolation switch connecting the secondary circuit to the second end of the first isolation switch, the control circuit being configured to:

control the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second and third isolation switches are closed or the first isolation switch is closed and the second and third isolation switches are open, and

control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second and third isolation switches are closed.

12. Arc furnace facility comprising:

an arc furnace including:

at least one electrode, and

a power supply system according to claim 10,

wherein the electrode is connected to at least one phase of the secondary circuit of the polyphase transformer.

13. The arc furnace facility according to claim 12, wherein the arc furnace comprises a plurality of electrodes, each electrode being connected to a different phase of the secondary circuit.

14. The arc furnace facility according to claim 12, wherein the power supply system further comprises a rectifier connected to the electrode (250a) and to each phase of the secondary circuit to supply the electrode with a continuous voltage from the secondary circuit.

15. A method for controlling a power supply system of an arc furnace, the power supply system including:

a power converter comprising an input connected to a polyphase supply grid (G) and an output, and

a polyphase transformer comprising a primary circuit and a secondary circuit,

a first disconnecting means, and

a second disconnecting means,

the method comprising:

controlling the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, a first isolation switch

is open and a second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and

controlling the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed,

wherein the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch.

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